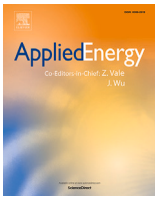
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Review of deep learning techniques applications in modern power systems stability and cyber security

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HIGHLIGHTS

- The paper's novelty lies in its dual focus on methodological detail and integrated analysis of stability and cybersecurity challenges—something few prior reviews achieve in such depth. Rather than simply listing existing methods, the paper groups them into meaningful categories and offers critical insights—not only examining past developments, but also current methodological limitations and proposing design considerations for future AI models.
- The paper includes detailed tables highlighting input features, model architectures, training strategies, and evaluation metrics, which enhance the readers' ability to compare methods across studies.
- The paper provides graphical representations of core research concepts central to dynamic- and cyber-physical phenomena, coupled with a classification logic that clarifies conceptual categories and methodological approaches within the reviewed literature, thereby offering a structured roadmap to guide future research and identify knowledge gaps.
- The paper integrates tables that systematically present the intrinsic advantages of each technique, based on their core characteristics. This enables clearer differentiation of models beyond application context, supporting more informed decision-making in addressing power system stability and cybersecurity challenges.
- The paper proposes innovative applications and interdisciplinary approaches that could improve the application of advanced AI methods in power systems.

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Cyberattack mitigation

ABSTRACT

The integration of renewable energy sources and modern technologies into power systems has introduced significant challenges related to stability, security, and control. This paper presents a comprehensive review of advanced artificial intelligence—specifically deep learning techniques (DLTs)—used to address these challenges, focusing on two main areas: (i) dynamic phenomena (such as small-signal, transient, voltage, frequency, and converter-driven stability) and (ii) cyber-physical phenomena (such as intrusion detection, false data injection, and cyberattack mitigation). Unlike previous reviews, this study goes beyond listing methods by grouping them meaningfully, discussing challenges, and proposing future design considerations. The novelty of this review lies in its dual thematic structure and detailed discussion of innovative applications of DLTs in power systems, their performance relative to other methods, and their application-specific and intrinsic advantages. It also provides a critical analysis of input features, model architectures, training strategies, and evaluation metrics. Moreover, it offers valuable insights not only into what has been done, but also into where current approaches fall short (e.g., in adaptivity, scalability, data security, and data efficiency) and how future research should progress. Additionally, the paper presents core concepts central to current research and connects them to specific methods via the application logic of DLTs, such as classification, prediction, detection, and control. By addressing current limitations and exploring future design recommendations, including interdisciplinary approaches, this review aims to support researchers and practitioners in using DLTs to develop stable, secure and efficient power systems of the future.

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Nomenclature

List of Acronyms

ACT	Average Computation Time.
ADAI	Average DSA accuracy index.
ART	Average response time.
ATSA	Average time to generate sample attack.
AVE	Average voltage deviation.
BPNN	Backpropagation neural network.
CRPS	Continuous ranked probability score.
CT	Compute time.
DAE	Deep auto-encoder.
DNN	Deep neural network.
DR	Detection rate.
DT	Decission tree.
ELM	Extreme learning machine.
FPR	False prediction rate.
GGAT	Global graph attention network.
GP	Non-parametric Gaussian process
IL	Imbalance learning.
IPLS	Improved power loop scheme.
KNN	K-nearest neighbor.
LHS	Latin hypercube sampling.
MA-ACE	Mean absolute ACE.
ME	Misclassification error.
MGC	Mean Grid cost.
MI	Mutual information.
MLP	Multilayer perceptron.
MPC	Model predictive control.
NEC	Number of episode before convergence.
NN	Neural network.
NN-DR	NN based density regularization technique.
p-loss	Pinball loss.
PCE	Polynomial Chaos Expansion.
PSO	Particle swarm optimization.
PT	Prediction time.
RBFNN	Radial basis function NN.
REL	Robust Ensemble Learning.
RF	Random forest.
RVFL	Random vector functional link
SDAE	Stacked-denoising auto-encoders.
SGPR	Sparse Gaussian process regression.
SVM	Polynomial kernel support vector machine.
TA_GRU	Time-adaptive GRU.
TF	Transfer function.

1. Introduction

Traditional power systems operations and dynamic characteristics are undergoing profound transformation due to factors such as transportation electrification, large-scale integration of inverter-based renewable sources (IBRS) and utilization of advanced communication devices. While these changes enable greater sustainability and efficiency, they also pose significant operational challenges [1–3]. For instance, IBRS's intermittency combined with variability of demand side may result in a highly volatile net load profile, which is a challenge for a system operator to balance [4,5]. Moreover, replacing conventional generators with IBRS significantly reduces system inertia [6,7]. This reduction in inertia, in turn, can cause the power system to experience a spectrum of operational challenges, such as large frequency excursions [8–11], and transient stability issues [12–15]. Addressing these requires fast-acting control mechanisms, such as synthetic inertia and fast frequency response provided by smart inverters [16]. In addition, IBRS, particularly distributed energy resources, are often interconnected, interoperable, and remotely controlled, making safeguarding of their cybersecurity critically essential. As highlighted in [17], the dependence of IBRS on communication networks and fast-acting digital controllers increases the attack surface, thus widening the cybersecurity risks associated with modern power systems. For example, although synthetic inertia can improve frequency stability, its implementation through software and communication protocols makes it vulnerable to cyber threats. A compromised inverter within an IBRS setup could, for example, inject erroneous or delayed inertial responses, thereby exacerbating frequency excursions rather than mitigating them, with direct impacts on overall grid stability.

Furthermore, modern power systems are intricately interdependent cyber-physical systems. The growing penetration of large-scale IBRS and increasing reliance on communication networks mean that the physical grid dynamics are closely coupled to the digital systems that control them. This interdependence between the physical and cyber layers of the power grid creates a tightly coupled cyber-physical system, wherein vulnerabilities in one domain can directly affect the other. For instance, cyberattacks such as false sensor data injection or denial-of-service attacks targeting a controller or communication link can cause delayed or incorrect control actions, potentially triggering voltage or frequency instabilities. Conversely, unstable physical conditions, such as fast frequency swings in a low-inertia power system, can reduce the reliability of cybersecurity monitoring tools, increase false positives in anomaly detection, or affect real-time decision support systems. The proliferation of distributed energy resources and power-electronic controllers further broadens these vulnerabilities, as each new additional inverter-energy source or communication link is a potential point of attack. In such a cyber-physical grid, malicious actions in the digital layer can translate into real physical phenomena, such as sustained oscillations or loss of synchronism, jeopardizing system stability. [18–20]. Therefore, remediating these intertwined and evolving challenges requires innovative assessment and control techniques.

Traditionally, power grid analysis is done using time-domain simulation (TDS). However, as the system expands with the proliferation of distributed IBRs, TDS becomes computationally expensive and inefficient for real-time assessment. This inefficiency extends to the detection and mitigation of cybersecurity threats, where the rapid analysis of system vulnerabilities is critical. Thus, conventional methods struggle to meet the demands of real-time stability assessment and robust cyber defense in the modern power system.

In response to these challenges, artificial intelligence (AI) methods, which include machine learning (ML) [21,22] and deep learning approaches [23], offer a promising alternative. Advanced AI techniques (i.e., DLTs) are well suited for analyzing large and complex datasets in real time. Compared to traditional AI techniques such as expert systems and shallow learning models—which are often constrained by static models, centralized data processing, feature-engineered datasets, predefined rule sets, and limited adaptability to real-time disruptions—DLTs offer a transformative alternative. Specifically, DLTs provide adaptive, scalable, and context-aware mechanisms capable of capturing the high-dimensional, nonlinear dynamics and time-varying characteristics of modern cyber-physical power systems. Moreover, these techniques can enable the concurrent modeling of physical system dynamics and cyber interaction patterns, which facilitates proactive responses to simultaneous threats to both system stability and cybersecurity. For example, deep reinforcement learning agents can learn real-time corrective control actions that adapt to both system disturbances and malicious intrusions. Likewise, graph neural networks can model the topological evolution of interconnected systems under

Table 1
Comparative analysis of our paper and recent reviews on power systems.

Feature / Criteria	Our Work	[27]: Year 2024	[23]:Year 2021	[30]: Year 2019	[28]: Year 2023	[29]: Year 2022
Dual Focus: Stability & Cyber-Physical security	Yes	No	No	Partially	No	No
Focus on power system Stability as a central topic	Yes	No	Yes	Partially	Yes	Partially
Focus on Cyber Security	Yes	No	No	Partially	No	No
Thematic division of AI- application (e.g., control vs prediction)	Yes	Partially	Partially	No	Partially	Yes
Challenges or limitations discussed	Yes	Yes	Yes	Yes	Yes	Yes
Power system concepts linked to specific AI methods	Yes	No	Partially	No	Yes	No
Comparison of Inputs,models, metrics, & advantage	Yes	Partially	Yes	Partially	Yes	No
Discussed intrinsic Method Advantages	Yes	No	Yes	Yes	Yes	Yes
Proposed Future Design Guidelines	Yes	Partially	Yes	Yes	Yes	Yes
Method Types (e.g., time-series, spatial, hybrid)	Yes	Yes	Yes	No	Yes	Yes

physical contingencies and cyberattacks. In addition, transfer learning and federated learning approaches promote knowledge sharing across geographically distributed assets without compromising data privacy, thereby supporting collaborative and cyber-resilient control strategies. Importantly, DLTs transcend the siloed application of traditional AI by integrating cyber situational awareness with physical system state estimation in real time. This unified framework not only improves the accuracy of threat detection and mitigation but also strengthens overall system resilience by adapting to novel attack vectors and evolving operational conditions. Such integrated capabilities are increasingly vital in modern power grids, where the boundary between physical stability and cybersecurity is progressively blurring, necessitating an intelligence-driven approach that DLTs support [2,17,24,25].

Recent reviews have explored various ML methods applied to power system challenges. For example, traditional ML techniques have been reviewed for applications like cyberattack detection and power quality [21,26]. Other reviews have focused on specific domains, such as stability assessment, covering aspects like frequency and voltage stability [2]. In addition, Khodayar et al. [23] reviewed three categories of deep learning for power system research and compared their variations. Prior reviews have primarily examined AI methods in domains such as computational electromagnetics and power system component modeling [27], stability assessment [23,28,29] and operational control, while only briefly mentioning cybersecurity [30]. However, most recent reviews focus on AI methods for power system stability, with limited or no attention to cybersecurity. In contrast, our work contributes by providing dual thematic treatment of power system stability and cybersecurity—areas that have often been treated separately in previous reviews. Table 1 offers a comparative overview of related surveys, illustrating the unique contributions and novel insights of our review. It shows that a gap still exists in current reviews on the integrated application of deep learning techniques to address the interconnected challenges of power system stability and cybersecurity in modern power grids.

Therefore, this paper critically conducts a detailed review of recent research on applications of advanced AI techniques in the last 10 years, with a particular focus on deep learning techniques (DLTs) aimed at improving the stability and security of power grids. Specifically, it reports the implementation of cutting-edge approaches such as transformer-based models, attention mechanisms, generative adversarial networks-based models, self-supervised learning, deep neural architectures, deep reinforcement learning-based models, and deep hybrid AI methods. These advanced methodologies, which extend beyond traditional supervised ML models, are referred to as DLTs throughout this work. In particular, this paper is structured around two principal categories: dynamic phenomena and cyber-physical phenomena, which are analyzed and grouped distinctly. Unlike previous reviews, this work not only presents state-of-the-art techniques but also discusses their limitations, identifies areas for potential improvement, and outlines promising directions for future research.

The deliberate selection of power system stability and cybersecurity as the core focus of this paper stems from their increasing complexity, interdependence, and the pressing need for intelligent, adaptive, and real-time decision-making capabilities in modern power systems. Although power system stability and cybersecurity address distinct problem areas, namely physical dynamics versus digital threats, they share methodological commonalities in the application of AI. For instance, techniques such as deep learning and reinforcement learning are widely utilized in both domains for tasks such as anomaly detection, pattern recognition, and adaptive control. To illustrate, predictive models support both transient stability assessment and intrusion detection, while reinforcement learning facilitates the development of optimal control strategies and robust cyber response mechanisms. Nevertheless, despite these methodological overlaps, each domain presents unique challenges that require specialized AI applications. On one hand, stability applications focus on dynamic system behavior, such as frequency regulation, voltage control, and rotor angle stability, which often require real-time, model-informed AI techniques. On the other hand, cybersecurity involves identifying and mitigating threats like data manipulation, malware propagation, and unauthorized access, tasks that rely heavily on adaptive, data-centric AI models capable of working effectively in adversarial environments. In addition, what sets these two domains apart from other areas of AI application in power systems, such as economic dispatch, renewable energy integration, or load forecasting [26,31], is their demand for high-stakes real-time decision making. Whereas many AI-driven solutions in economic dispatch or renewable energy forecasting center on optimization and predictive modeling for cost and efficiency improvements, applications in stability and cybersecurity are often mission-critical [21,23]. Consequently, they require low-latency decision-making and adaptive learning capabilities to mitigate cascading failures or cyberattacks in real time. By examining power system stability and cybersecurity in tandem, this review highlights how advanced AI methods are being utilized to improve both the physical and cyber resilience of power systems. Ultimately, this review aims to present perspectives on current deep learning applications in power stability and power cybersecurity, while also encouraging cross-domain innovation and integrated approaches to modern power system challenges.

The major contributions of this paper are:

- The paper's novelty lies in its dual focus on methodological detail and integrated analysis of stability and cybersecurity challenges—something few prior reviews achieve in such depth. Rather than simply listing existing methods, the paper groups them into meaningful categories and offers critical insights—not only examining past developments, but also current methodological limitations and proposing design considerations for future AI models.
- The paper includes detailed tables highlighting input features, model architectures, training strategies, and evaluation metrics, which enhance the readers' ability to compare methods across studies.
- The paper provides graphical representations of core research concepts central to dynamic- and cyber-physical phenomena, coupled with a classification logic that clarifies conceptual categories and methodological approaches within the reviewed literature, thereby

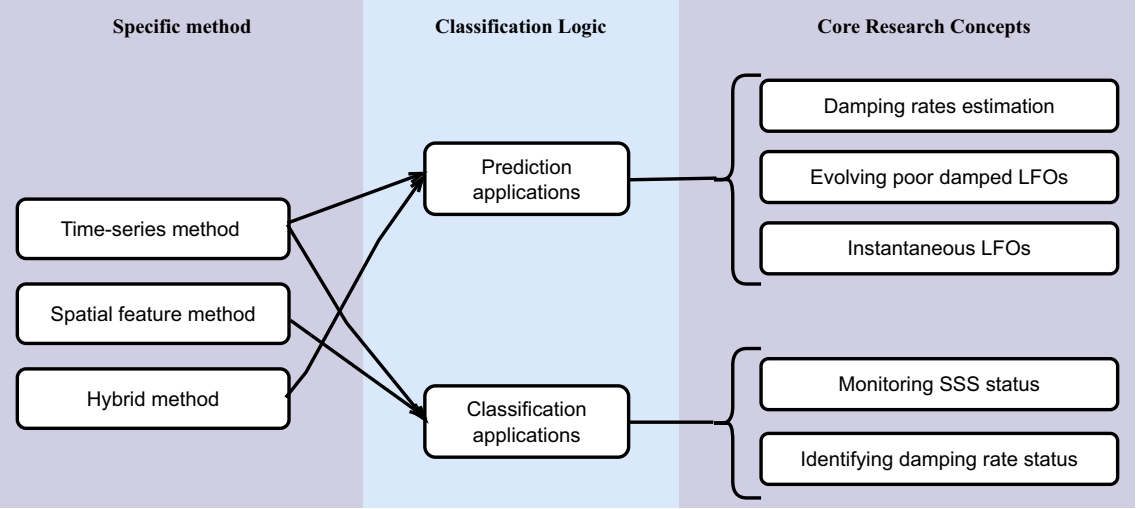


Fig. 1. The application of dlts for power systems small signal stability: core concepts central to the present research.

offering a structured roadmap to guide future research and identify knowledge gaps.

- The paper integrates tables that systematically present the intrinsic advantages of each technique, based on their core characteristics. This enables clearer differentiation of models beyond application context, supporting more informed decision-making in addressing power system stability and cybersecurity challenges.
- The paper presents the intertwined phenomena of the two domains, discussing scenarios in which cyberattacks can trigger instability and thereby underscore their interconnectedness.
- The paper proposes innovative applications and interdisciplinary approaches that could improve the application of advanced AI methods in power systems.

Figs. 1, 3, 5, 7, and 9 present graphical highlights of the core research concepts of current DLTs applications in power system stability, while Figs. 11, 13, and 15 represent those related to cybersecurity. The content of each section of this paper is structured into two distinct groups: (1) deep reinforcement learning (DRL)-based control problems, and (2) deep learning-based prediction, classification, or detection problems. The remainder of this paper is as follows. Section 2 presents the current applications of DLTs in several areas of power network stability evaluation, prediction, and control, including discussions of design challenges. In Section 3, the paper reviews DLTs studies on power system security problems: detection, creation, and mitigation, including discussions of design challenges. In Section 4, the paper discusses the interdependence between cybersecurity and stability. Section 5 presents suggestions for future research directions. Finally, Section 6 concludes the paper.

NOTICE: Sections 2 and 3 discuss the independent application of deep learning techniques for power system stability and cybersecurity, with minimal overlap. The separation of Sections 2 and 3 is intended to reflect current research trends, where integrated DLT-based approaches that simultaneously address both stability and cybersecurity challenges are rare. Apart from Reference [32], which employs federated learning for decentralized inertia estimation within a privacy-preserving client–server framework, most existing studies investigate DLTs' solutions that treat both domains independently. The scarcity of such cross-domain DLTS applications underscores a novel gap in the literature despite the inherent interdependence of stability and cybersecurity in modern cyber–physical power systems. Therefore, by providing a domain-specific overview, this review creates a solid knowledge base from which future cross-domain DLTs' innovations can develop and promote integrated frameworks that simultaneously address both stability and security challenges in modern power systems.

2. Power system stability enhancement using deep learning-based techniques

2.1. Small signal stability

A power system is said to exhibit small signal stability (SSS) when it maintains synchronism under minor perturbations (such as changes in load or generation). However, inadequate damping torque in the system can significantly contribute to SSS challenges and the occurrence of poorly damped low-frequency oscillation modes (LFOs). These oscillations can limit power transfer capabilities and, in extreme cases, cause system-wide instability. In response to these challenges, recent research has demonstrated the efficacy of various DLTs in addressing LFOs challenges. However, the existing literature primarily focuses on prediction- and classification-based applications. Fig. 1 encapsulates the core concepts central to the present research.

In terms of **deep learning-based prediction applications**, Muhammed et al. [33] proposed deep learning methods for accurate real-time forecasting of evolving poorly damped LFOs and reconstruction of damping rates in power systems integrated with wind power plants. This was achieved through efficiently trained stacked long-short-term memory (LSTM), convolutional neural network (CNN)-LSTM and convolutional LSTM (Conv-LSTM) models. Based on the evolution of LFOs, Azman et al. [34] developed an LSTM model to predict LFOs after disturbances. Similarly, Satheesh et al. [36] trained LSTM recurrent neural network (LSTM-RNN) to estimate instantaneous low-frequency oscillatory modes of a power system.

Furthermore, in the domain of **deep learning-based classification applications**, [35] designed a CNN model to monitor small-signal status of power system. Within the CNN model, an auto-encoder (AE) and a generative adversarial inception network (GAIN) are incorporated to facilitate feature selection and to enhance robustness against missing data or partial observability. In addition, Cepeda et al. [37] trained RNN model to identify the status of damping rates of inter-area oscillation modes. The real-time estimation of oscillation modes is conducted using ELPROS, which utilizes a proprietary modal identification algorithm. ELPROS is applied to gather active power data from PMUs. Similarly, Rojas et al. [38] investigated five deep learning architectures: multilayer perceptron, fully-convolutional neural network, inception network, time convolutional NN, and multi-channel deep convolutional NN to classify small-signal stability. Table 2 presents the critical information in the current literature on DLT-based SSS studies, and Table 3 summarizes the intrinsic advantages of the proposed DLTs from SSS studies.

Table 2
The critical information based deep learning techniques for sss, low-frequency, damping rates prediction and reconstruction.

Ref. No	Method	Model input features	Model output	Training strategies	Methods comparison, performance meterics and range of accuracy	Advantages of proposed method
[33]	LSTM, CNN-LSTM, Conv-LSTM	Bus voltage magnitude & voltage angle	Low-frequency oscillatory modes; damping rates	- Model check-point; - Early-stopping - Adam optimizer	1. Stack LSTM | **RMSE: 0.00025 - 0.00266**; **MAAPE: 0.156 - 2.672** 2. CNN-LSTM | RMSE: 0.00050 - 0.00231; MAAPE: 0.335 - 2.812 3. Conv-LSTM | RMSE: 0.00046 - 0.00290; MAAPE: 0.352 - 3.318 4. SSLSTM | RMSE: 0.00065 - 0.01140; MAAPE: 0.745 - 12.041	Effectively predicts small-signal stability (SSS) in renewable-integrated power grids
[34]	LSTM	Bus voltage phasors	LFOs; damping ratios	- Adam optimizer - Learning rate decay - Use stable transients data	1. LSTM-64 | RMSE: 0.0008 - 0.01234; MAAPE: 0.9317 % - 14.5678 % 2. LSTM128 | RMSE: 0.00068 - 0.01140; MAAPE: 0.7883 % - 12.0411 % 3. LSTM-256 | RMSE: 0.00065 - 0.0119; MAAPE: 0.745 % - 13.9556 %	Enables accurate prediction of SSS response over extended time steps
[35]	AE + GAIN + CNN	Active - & reactive powers	Secured or unsecured status	- Feature selection - handling missing data - Random search - Adam optimizer	1. Proposed | **ACC: 99.07 % - 99.21 %** 2. RF + CNN | ACC: 97.89 % - 98.22 % 3. Relief + CNN | ACC: 95.55 % - 95.92 % 4. Mutual info + CNN | ACC: 96.89 % - 97.52 %	Improved PMU interpretability and feature selection via GAIN and AE.
[36]	LSTM-RNN	System voltage	Oscillatory amplitude, frequency, damping ratio	- PMU data preprocessing - Variational Mode Decomposition (VMD) - Teager Kaiser Energy Operator (TKEO)	1. LSTM-RNN | **MAE: 0.001533**; **MAPE: 8.58 %** 2. Backprob NN (BPNN) | MAE: 0.1956; MAPE: 26.85 % 3. RBFNN | MAE: 0.136; MAPE: 22.57 % 4. GRU | MAE: 0.152; MAPE: 18.63 %	Faster and more adaptable than existing methods.
[37]	RNN	Active power: damping, amplitude frequency	Damping ratio	- Dropout; - Early stopping	|	Enables oscillatory stability forecasting using WAProtector streaming data.
[38]	FCN, MCDCNN Inception, MLP, CNN	Line current; bus voltage	Stable/unstable labels;	- k-fold cross-validation	1. FCN | predict time (PT): 1.20 - 1.21 ms 2. Inception | PT: 2.43 ms - 2.50 ms 3. MCDCNN | PT: 0.477 ms - 0.491 ms 4. MLP | PT: 0.225 ms - 0.235 ms 5. CNN | 0.261 ms - 2.75 ms	MCDCNN exhibit optimal performance

Table 3
Intrinsic advantages of proposed dlts from small signal stability studies.

Ref. No	Method	Intrinsic advantage
[33,34,39]	LSTM	Effective modeling of long-term temporal dependencies in sequential data through gated memory, avoiding vanishing gradients.
[33,40,41]	CNN-LSTM	Combines spatial feature extraction with temporal modeling, enabling efficient learning with fewer parameters and reduced overfitting.
[33]	Conv-LSTM	Incorporates spatial structure directly into temporal state transitions, enabling high-dimensional spatiotemporal pattern learning.
[35]	AE	Reduces data dimensionality while preserving interpretability, improving learning efficiency.
	GAIN	Robustly imputes missing data with statistically consistent values, enabling reliable analysis under partial observability.
[35,38,42,43]	CNN	Extracts complex spatial features from 2D-structured input.
[36,44]	LSTM-RNN	Superior capability to model long-term temporal dependencies without suffering from vanishing gradients, unlike traditional RNNs
[37]	RNN	Inherently capture temporal dependencies in sequential data through internal memory loops, enabling short-term forecasting.
[38]	MCDCNN	Parallel convolution across multiple dimensions of input enables efficient feature extraction with minimal preprocessing.
	Inception	Utilizes parallel convolutional filters of varying sizes for capturing multi-scale temporal patterns.
	FCN	Employs global average pooling and deep convolutional layers to extract hierarchical features.
[38,45–47]	MLP	A fully-connected structure that serves as a baseline model; low-latency suitable for shallow patter recognition.

Fig. 2 clusters existing schemes based on input features and output configurations.

2.1.1. Challenges in current DLTs designs for SSS forecast

The methods discussed in section 2.1, while innovative and effective in many scenarios, have several limitations. Firstly, a key drawback lies in their inability to effectively capture abrupt and inconsistent oscillatory patterns. Although the approaches outperform traditional models in terms of accuracy, [33] highlights specific cases where they fail to track sudden changes in oscillatory modes, particularly in systems with high penetration of renewable energy. This is because the stochastic nature of such systems introduces dynamic variations that these models cannot interpret. In addition, when focusing on limited accuracy in predicting small-signal stability, [34] notes that LSTM often underestimates or overestimates eigenvalue components, especially during rapid fluctuations. Consequently, this can result in incorrect stability assessments and inappropriate control decisions. Another limitation is the computational intensity of methods such as Conv-LSTM, CNN-LSTM, and LSTM-RNN [33,37], which involve numerous trainable parameters. These parameters necessitate considerable computational resources and long training times. For instance, training a single architecture on a GPU typically averages around six hours, rendering these methods unsuitable for time-critical applications or for systems with limited computing resources.

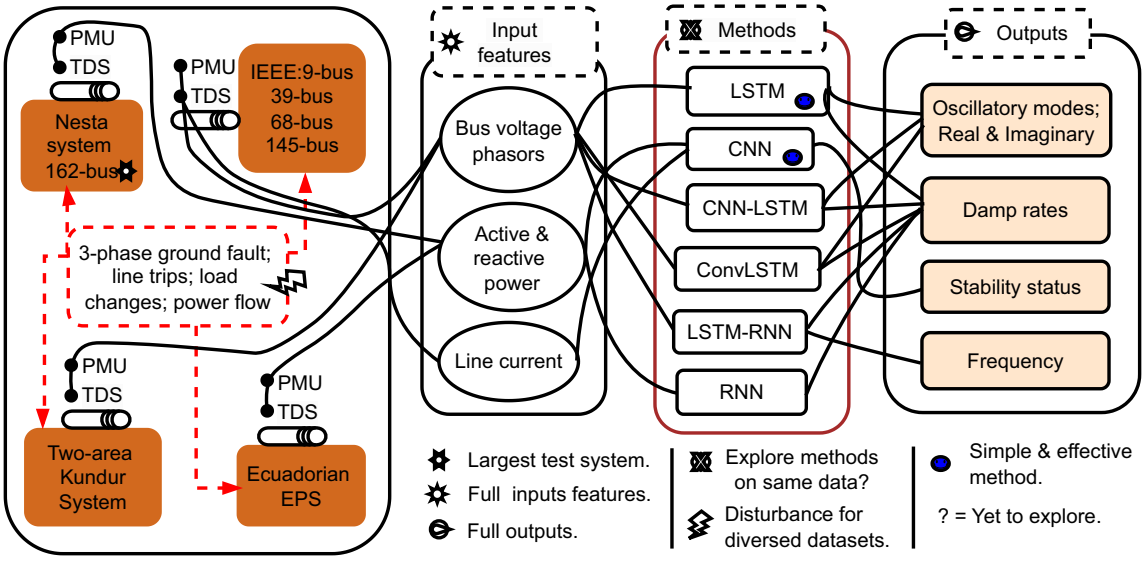


Fig. 2. Insightful framework for robust practical DLT's forecasting model development for SSS evaluation and for further research consideration.

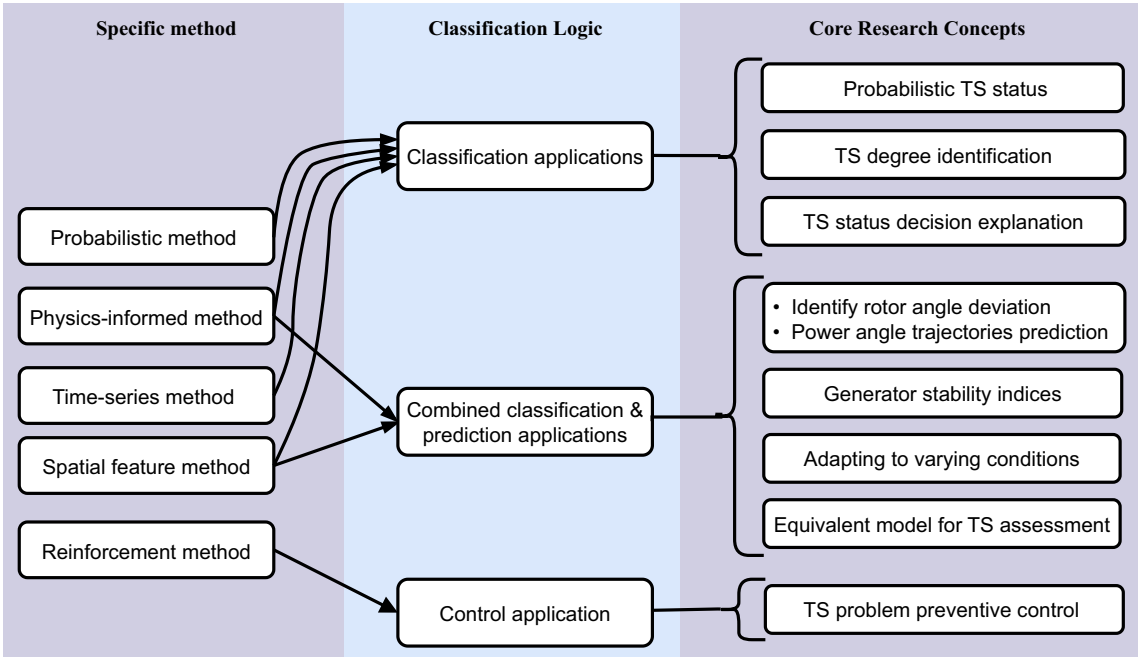


Fig. 3. The application of dlts for power systems transient stability: core concepts central to the present research.

Additionally, generative adversarial imputation networks (GAIN), as utilized in [35] to address missing data, perform reliably with up to 20 % data loss. However, their effectiveness is contingent on the distribution and proportion of missing values, and they can underperform in extreme scenarios such as widespread communication failures or significantly higher data loss rates. Moreover, their reliance on imputed data introduces additional uncertainty, which can compromise predictive accuracy in critical settings. Furthermore, preprocessing steps, such as noise filtering, detrending, and window framing, as discussed in [36], add extra layers of complexity. These steps are prone to errors that can undermine prediction performance if not executed accurately. Lastly, while the methods provide predictive insight, they are not directly integrated with automated control mechanisms, such as adaptive relays or flexible AC transmission systems (FACTS), that proactively mitigate oscillatory risks.

2.2. Rotor angle stability

Rotor angle stability, also called transient stability (TS), is critical for maintaining synchronous operation of generators during significant disturbances. Failure to ensure rotor angle stability can result in widespread outages and loss of system integrity. In recent years, research has increasingly focused on developing deep learning techniques (DLTs) that can handle the high-dimensional and nonlinear nature of transient stability problems. Accordingly, this section categorizes the existing literature into three main segments: deep learning-based classification applications, combined classification- and prediction-based applications, and deep reinforcement learning-based control applications. Fig. 3 illustrates the core concepts underlying the present research.

To begin with, in the context of **deep learning-based classification applications**, Ye et al. [14] addressed the challenge of uncertainties

arising from inverter-based resources and stochastic loads. Specifically, the authors proposed a physics-informed sparse Gaussian process (PISGP), integrated with stochastic variational inference, to quantify probabilistic TS status and rotor angles for power systems with uncertain dynamic photovoltaic (PV) power plants and loads. Similarly, Tan et al. [13] utilized neural networks with a transient stability index (TSI) density weighting scheme and a regularization technique (NN-DR) to quantify probabilistic TS status from unpredictable wind generation and load variations.

Furthermore, Zheng et al. [49] developed a structured reserve deep belief network (SR_DBN) for assessing TS status. Likewise, Li et al. [50] devised an active transfer learning method with DBN (ATL-DBN). The DBN is first trained as a base model for better TS status and TS degree assessment. When the operating conditions change significantly, minimum number of informative samples is selected using active learning scheme and maximum mean discrepancy (MMD) to determine appropriate transfer learning strategy. In addition, Azman et al. [34] devised a CNN method to identify TS status of rotor angle during disturbances using system voltage phasors. Moreover, leveraging an explainable DLT framework, Chen et al. [55] proposed a time-adaptive model based on a dual-stage attention gated recurrent unit (DSA-GRU) for TS assessment (TS_A). This architecture includes a feature block and a time-attention block to interpret the patterns learned by the TS_A model. Time-adaptive TS_A is achieved through multiple channels that handle the features at various stages. In contrast, Li et al. [56] introduced a continual learning approach for monitoring TS status. By using an improved CNN-based orthogonal weight modification (OWM) algorithm, the model maintains accuracy by continually learning from new data during online prediction without retraining the model from scratch.

Moving on to **combined classification and prediction based applications**, Li et al. [48] developed a physics-informed deep learning (PIDL) model to evaluate after disturbance system's TS. A trained RNN model is then used to forecast rotor speed deviation $\Delta\ \omega_i$ and power angle trajectories ($\Delta\ \delta_i$). Likewise, An et al. [51] constructed a two component CNN scheme to evaluate TS margin of power system based on security region under anticipated contingencies. The first component is utilized to classify input samples into stable, unstable or uncertain classes. The second component is used to forecast the stability indices of generators for the classified samples. Similarly, Shi et al. [52] developed a bidirectional active transfer learning (Bi-ATL) deep learning model for adaptive TS assessment and dominant instability mode (DIM) identification. The Bi-ATL scheme integrates three sets of data from instances and a model of the initial operating condition, along with a few labeled data points from the new operating state to enhance adaptive efficiency. In addition, [53] trained a series of deep LSTM networks to construct a dynamic equivalent model (DEM) to represent the TS of hybrid renewable energy sources (HRES), using an unsupervised clustering algorithm to categorize power responses over time. Similarly, Zhu et al. [54] developed a hierarchical deep learning machine (HDLM) based on a two-level CNN regression model to achieve both quantitative and qualitative online prediction of TS efficiently.

Finally, in the realm of **deep reinforcement learning-based control**, Zeng et al. [57] developed a deep reinforcement learning (DRL) approach to serve as a preventive TS control scheme with lower computational cost. In addition, Chen et al. [58] devised a control approach based on DRL agent for improving the TS of a multi-virtual synchronous generator system (MVS) that incorporates resistive superconducting fault current limiters (RSFCL) coupled to a weak power grid. Tables 4 and 5 present the critical information in the current literature on DLT-based TS studies, while Table 6 summarizes the intrinsic advantages of the proposed DLTs from TS studies. Fig. 4 groups existing concepts based on input features and output configurations.

2.2.1. Challenges in current DLTs designs for TS assessment

The reviewed DLTs present innovative approaches for improving TS_A, yet they come with several limitations that could challenge their real-world applicability. Firstly, the method in [13], relies on constructing pairwise similarity matrices and implementing kernel density estimation; consequently, its computational burden can severely hinder scalability for large datasets or high-dimensional systems. Moreover, the framework in [14] depends on high-quality, physics-informed training data, which presupposes accurate system models that might not always be attainable. Similarly, PIDL [48], while effective, employs the forward Euler method for swing equation discretization, which may introduce numerical inaccuracies compared to more advanced higher-order methods. In addition, the regularization strategy incorporating grid structure [49] improves interpretability but simultaneously increases parameter-tuning complexity and depends on accurate grid modeling. Furthermore, the fine-tuning strategies guided by calculated MMD in [50], prioritize computational efficiency but overlook the potential benefits of selectively tuning intermediate layers. Likewise, the accuracy of computer-vision-based power flow images in [51] depends on precise preprocessing and up-to-date topology, both of which may not always be guaranteed. In the Bi-ATL approach in [52], the backward active-learning step significantly depends on the quality of source-domain data; thus, noisy or irrelevant samples can undermine the dataset and degrade model performance. Monte Carlo simulations used in [53] for HRES plant modeling rely on assumed uncertainty ranges that may limit the generalizability of model in diverse operating conditions. Graphical transient characterization in [54] is highly sensitive to noisy or incomplete data, posing challenges under sparse or inconsistent data measurements. Meanwhile, interpretability-guided optimization [55] necessitates manual adjustments when contingencies are mispredicted, making the process less scalable and potentially inconsistent across operators or systems. The orthogonal projection matrix in [56], while preserving prior knowledge, involves iterative updates that may not scale for highly dynamic or large-scale systems. Moreover, advanced neural network architectures with multiple agent workers in [57] demand significant memory and processing power, limiting their application in environments without high performance computing capabilities. Finally, the coordination between the DRL agent and RSFCL in [58] assumes consistent performance of the latter, however, factors such as temperature fluctuations or material aging can compromise their fault-current limiting behavior and thus affect the overall synergy of the strategy. These limitations collectively underscore the challenges in deploying these advanced techniques in practical, dynamic power systems.

2.3. Frequency stability

Frequency stability ensures the balance between generation and load under varying conditions [29,64]. The increasing integration of renewable energy sources, characterized by intermittency and variability, poses significant challenges to maintaining frequency stability. In response to these challenges, the development of DLTs for the forecasting and control of the frequency of the power system has received increasing research interest in recent years. Accordingly, this section categorizes the existing literature into two main areas: deep learning-based prediction applications and deep reinforcement learning-based control applications. Fig. 5 presents the core concepts that form the basis of current research.

In terms of **deep learning-based prediction applications**, some noteworthy contributions have emerged. For instance, Xie et al. [41] designed a combined CNN and LSTM network to predict the dynamic frequency of power systems. Their approach also provided an optimal event-based load shedding strategy following disturbances, thereby supporting frequency stability. Similarly, Wu et al. [60] used a multibranch ensemble full connection (MBEFC) network to effectively predict frequency response safety level and frequency response safety time of power system. In addition, Wen et al. [44] designed LSTM-RNN to accurately predict control reference signals, specifically, power fluctuations, from measured frequency data for

Table 4
The critical information based deep learning techniques for rotor angle stability.

Ref. Noe	Method	Model input features	Model output	Training strategies	Methods comparison, Performance meterics and range of accuracy	Advantages of proposed method
[13]	NN-DR	P_{wind}, P_{Load}, Q_{Load}, P_{SG}	TSI	Used loss function based on TSI weighting and similarity regularization	1. NN-DR | **MSE: 122.73 - 192.68** 2. NN without IL | MSE: 251.31 - 30,721 3. SGPR | MSE: 2431.27 - 3071.23	Enables fast and accurate TSA under imbalanced and complex data distributions.
[14]	PISGP	Uncertain PV power, load	TSI stability status; rotor angles	- Linearize DAE for regresion - Physics-informed samples - K-Means for data label - Stochastic variational inference method	1. PISGP | **MAPE: 3.14 - 5.69** 2. GP | MAPE: 3.18 - 8.34 3. LHS | MAPE: 4.12 - 10.34 4. PCE | MAPE: 7.32 - 49.78	High performance with low data and computation.
[48]	PIDL RNN	P_{m_i}, P_{e_i}; W_t	ΔW_t; $\Delta \delta_t$	- Transform continuous dynamics into a discretize time-series - Two layer RNN - L-BFGs optimizer	PIDL | Accuracy rate: 96.83 % - 100 %	Effective for transient angle stability assessment.
[49]	SRDBN	Active power, reactive power, bus voltage, voltage angle	Stable/ unstable status	- unsupervised pre-training - Supervised fine-tuning - loss func. with regularization term	1. SRDBN | **ME: 1.37 % - 1.80 %** 2. SVM | ME: 5.50 % - 6.55 % 3. Conventional DBN | ME: 2.63 % - 3.30 %	Highly interpretable and power system aware transient stability evaluation.
[50]	ATL-DBN	Power angle trajectories	Stable/ unstable labels; stability degree	- Pre-training & fine-tuning - Active learning for informative sample selection - Transfer learning via MMD	1. ATL-DBN | **accuracy (ACC): 98.92 %;RMSE: 0.036 - 0.049** 2. MLP | ACC: 96.26 %; RMSE: 0.245 - 0.458 3. Radial basis SVM | ACC: 96.83 %; RMSE: 0.229 - 0.466 4. DT | ACC: 97.18 %; RMSE: 0.042 - 0.073 5. RF | ACC: 97.85 %; RMSE: 0.059 - 0.110 6. KNN | ACC: 97.11 %; RMSE: –	Improves adaptive stability assessment in dynamic power system conditions.
[34]	CNN (GCNN, GINN)	Voltage phasor measurements	Rotor angle stability status	- Data augmentation - Dropout application - Nesterov stochastic gradient descent (SGDN)	1. GCNN | ART: 1.253 cycles 2. GINN | **ART: 1.106 cycles** 3. SVM | ART: 4 cycles 4. ELM | ART: 1.4 cycles 5. LSTM | ART: 1.448 cycles 6. CNN | ART: 1.111 cycles	Accurate and fast forecast of the transient stability status
[51]	CNN	Active power reactive power voltage images	generator stability indices	- Create powerflow images - Weighted cross-entropy to handle data imbalance - PSO to optimize CNN structure - Bp to fine-tune all parameters	1. CNN | **ACC: 97.33 %** 2. MLP | ACC: 95.25 % 3. SVM | ACC: 96.08 % 4. DT | ACC: 95.42 %	Improved margin detection using steady-state data.
[52]	1D-CNN	Rotor angle; voltage	Stable status Voltage instability rotor instability	- Bidirectional active transfer learning (Bi-ATL): Forward active learning; Backward active learning; Fine-tuning	1. 1D-CNN | ACC: **0.969 %** (0.9700 %) 2. LSTM (GRU) | ACC: 0.9575 % (0.9573 %) 3. ANN | ACC: 0.9718 % 4. XGBoost | ACC: 0.9752 % 5. RF (DT) | ACC: 0.9613 % (0.9557 %) 6. KNN (SVM) | ACC: 0.282 % (0.8453 %)	Improves model adaptability to varying operating conditions.
[53]	Deep LSTM-based DEM	Real & imaginary positive sequence voltage at PCC	Real & real power at PCC	- HRES plant production & SC line data, unsupervised clustering & PSO parameter tuning	1. LSTM-based DEMs | 0.04 %-to-0.3 % 2. TF-based models | smaller TSI error	Accurate time-domain power response in HRES TS assessment.
[54]	HDLM-based two-level CNN	Transformed (fault-on + 2) trajectories (V, δ, Δf, Q) transient 2D images	Stability margins	- Divide & conquer method - Transform data into 2D transient image - Rough CNN training - Refine CNN training - Adam optimizer	1. Proposed | **RMSE: 0.021; ACC: 99.25 %** 2. SVM | RMSE: 0.058; ACC: 93.08 % 3. DT | RMSE –; ACC: 96.42 % 4. RF | RMSE: –; ACC: 98.33 % ANN | RMSE: 0.047; ACC: 96.91 % 5. ELM | 0.069; ACC: 98.25 % 6. CNN | 0.024; ACC: 98.92 %	Reliable and fast online TSP with accurate stability status and margin prediction.
[55]	GRU	Voltage; phase angle	Stability status	- Sample labeling using TSI - Data normalization - Loss function & Adam optimizer - Model fine-tuning - Attention mechanism	1. Proposed | **F1-score: 0.963 - 0.985**; average decision time ($T_{d,avg}$/**cycles): 1.15** 2. TA-GRU | F1-score: 0.960 - 0.980; $T_{d,avg}$/cycles: 1.16 - 1.17 3. GRU | F1-score: 0.959 - 0.983; $T_{d,avg}$/cycles: 5.00 4. DT | F1-score: 0.922 - 0.946; $T_{d,avg}$/cycles: 5.00 5. SVM | F1-score: 0.943 - 0.961; $T_{d,avg}$/cycles: 5.00	Faster assessment with improved model interpretability.

Table 5
The critical information based deep learning techniques for rotor angle stability.

Ref. No	Method	Model input features	Model output	Training strategies	Methods comparison, performance meterics and range of accuracy	Advantages of proposed method
[56]	OWM-CNN	Voltage amplitude, voltage phase	Stable; unstable labels	- Orthogonal weight modification (OWM) algorithm - Continual learning	1. ANN | ACC: 96.07 % - 97.45 % 2. SVM | ACC: 96.21 % - 96.8 % 3. RF | ACC: 97.13 % - 97.58 % 4. CNN | ACC: 97.71 % - 98.36 % 5. OWM-CNN | **ACC: 97.48 % - 98.36 %**	Efficient updating with limited data and prior knowledge.
[57]	Distributed DDPG-based DRL	Generator & line active power; nodal phasor voltages	Adjusted output active power;adjusted terminal voltages	- Deep deterministic policy gradient (DDPG) - Markov Decision Process (MDP)	1. DRL | **MGC: $ 61,379.32 - $33,296.74; ACT: 1.0227s - 4.7681s** 2. TSCOPF | MGC: $61,084.66 - $ 327,051.68; ACT: 714.61s - 1255.43s	Allows safe and cost-efficient emergency transient stability control.
[58]	DRL|actor: CNN; Critic |GRU	Power vancacy P_{at}; output current I_t; angular frequency deviation W_{vsgt}; power angle δ_{vsgt}	Regulated reference power ΔP_{ref},regulated virtual inertia ΔJ	- Formulation the MDP - Use softmax DDPG(SD2) - Time series feature extraction network	1. Proposed | increases rotor angle stability 2. VSG control | loses rotor angle stability 3. VSG + RSFCL | loses rotor angle stability 4. DDPG | increases rotor angle stability 5. IPLS | loses rotor angle stability	Improves transient stability of MVS in unseen scenarios.

Table 6
Intrinsic advantages of proposed DLTs from rotor angle stability studies.

Ref. No	Method	Intrinsic advantage
[13]	NN-DR	Enhanced learning from rare events through density-based sample reweighting, improving model sensitivity to critical data regions.
[14]	PISGP	Removes the requirement for prior knowledge of uncertain input distributions, allowing for data-efficient learning in high-dimensional settings.
[48]	PIDL-RNN	Integrates physical laws into RNN architecture reduces the model's dependency on large training datasets.
[49]	SRDBN	Promotes similarity among electrically adjacent nodes, thus improving generalization.
[50]	ATL-DBN	It enables rapid model adaptation by selectively fine-tuning only relevant layers based on domain discrepancy, thus reducing computational cost.
[34,51,59]	CNN	Extracts complex spatial features from 2D-structured input.
[52]	1D-CNN	Excels at extracting localized temporal patterns in multivariate time series data while requiring fewer parameters compared to recurrent models.
[53]	Deep LSTM- based DEM	Captures long-term temporal dependencies in system dynamics without requiring detailed physical models.
[54]	HDLM-two level-CNN	Improves prediction correctness through hierarchical refinement via a divide-and-conquer strategy.
[55]	GRU	It enables efficient learning of temporal dependencies in time-series data with a simpler structure and lower computational cost than LSTM.
[56]	OWM-CNN	Enables continual learning by updating network weights without catastrophic forgetting, eliminating the need to retain or retrain on historical data.
[57]	Distributed DDPG based DRL	Improved training efficiency through parallelized experience collection and off-policy learning, greatly reduces convergence time compared to standard DDPG
[58]	DRL-actor: CNN; Critic-GRU	Improved spatiotemporal feature extraction for time-series inputs allows actor and critic networks to learn more robust and generalizable policies.

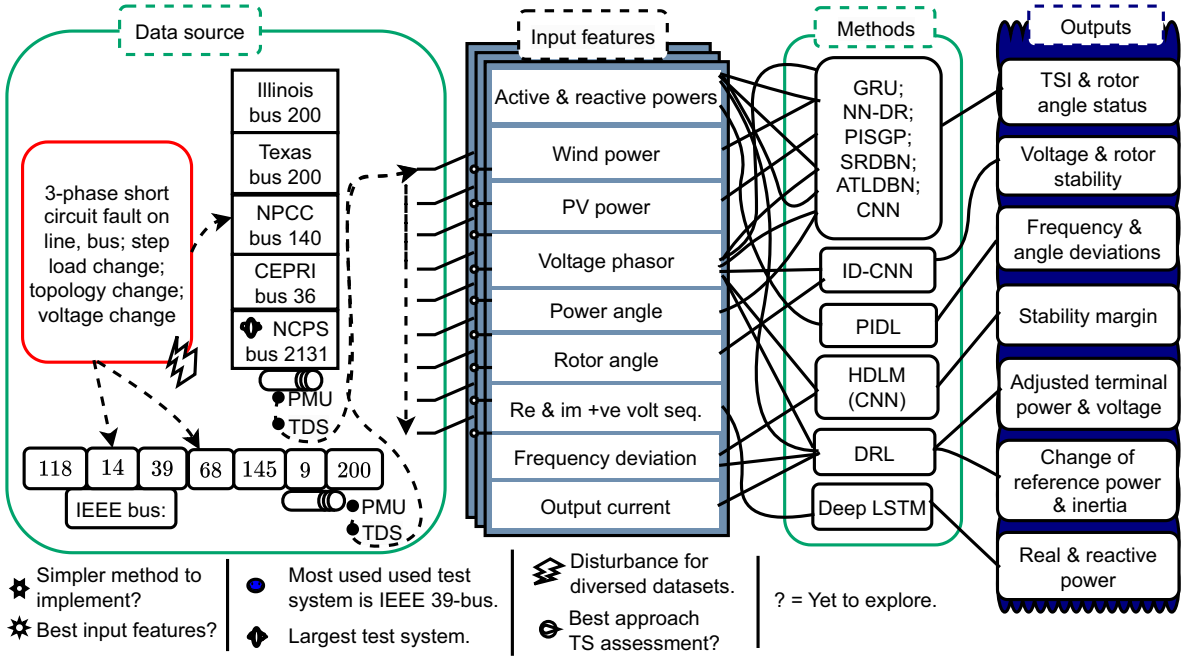


Fig. 4. Insightful framework for robust development of DLT's model for TS assessment and for further research consideration.

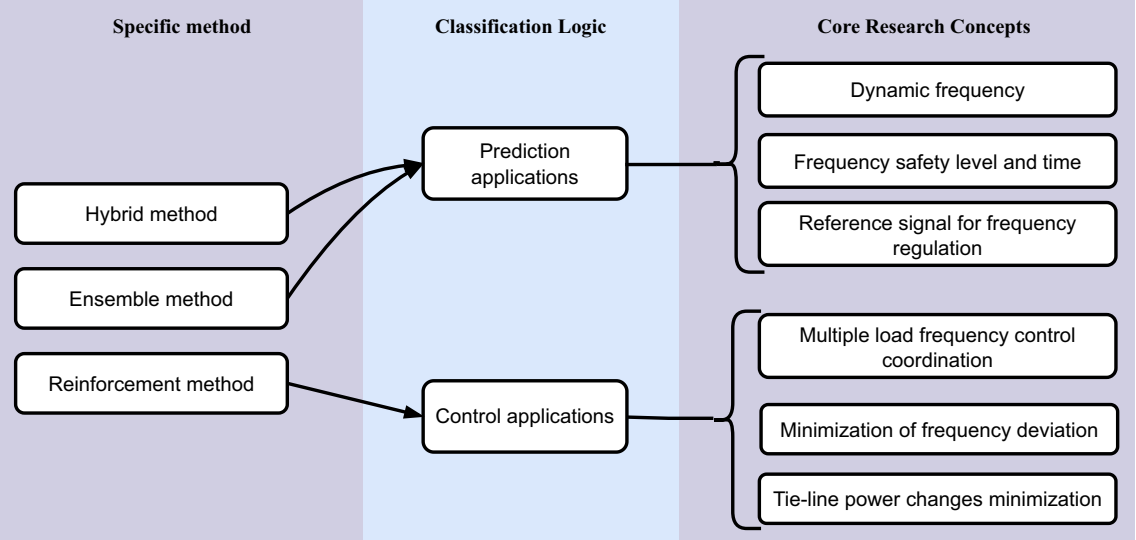


Fig. 5. The application of DLTs for power systems frequency stability: core concepts central to the current research.

use in load frequency control (LFC), to aid in the regulation of system frequency.

Turning to **deep reinforcement learning-based control applications**, good progress has also been made. For example, Yan et al. [61] developed a multi-agent deep reinforcement learning (MADRL) in a continuous action setting to optimally coordinate multiple load frequency controllers in a multi-area power system. The MADRL method operates as a centralized learning, based on a global action-value function, to effectively reduce frequency deviation and unexpected changes in tie-line power exchange. In a similar vein, Yang et al. [62] incorporated an attention mechanism within a critic network in multi-agent RL for load frequency control to minimize frequency deviation in interconnected power systems. Additionally, Khalid et al. [63] applied a twin-delayed deep deterministic policy gradient (TD3) RL agent to adjust the parameters of proportional-integral-derivative (PID) controllers for load frequency control. This approach not only minimizes deviations in both power system frequency and tie-line power, but also incorporates mechanisms within the TD3 framework to prevent negative gain values. The critical information in the current literature on DLTs-based frequency stability studies is presented in Table 7. Table 8 summarizes the intrinsic advantages of the proposed DLTs from frequency stability studies, and Fig. 6 groups existing concepts based on input characteristics and output configurations.

2.3.1. Challenges of DLTs designs for frequency stability

Although the current approach to frequency stability prediction and control using DLTs is undoubtedly innovative, it nonetheless encounters some limitations. For example, the process of building 4-D tensors in [41], which incorporates spatial-temporal feature extraction and dimensionality reduction, can become computationally intensive, particularly in systems with a large number of nodes or high temporal resolution requirements. Consequently, this computational demand poses challenges for real-time deployment where rapid predictions and immediate control actions are essential. Furthermore, the ensemble full connection structure in [60] lacks a dynamic weighting mechanism for integrating contributions from different branches, potentially resulting in suboptimal feature fusion. In addition, the method in [44] has yet to demonstrate robustness under noisy or incomplete data, or resilience to unexpected disturbances, leaving its reliability in practical scenarios uncertain. Moreover, the performance reported in [61] is significantly influenced by the quality of the initial PID controllers' database. Specifically, poorly tuned PID controllers or insufficient training data can result in suboptimal starting points, in turn prolonging convergence times, or yielding less effective control outcomes. Finally, the dependence on predefined parameters such as the reward scaling factor and soft update rate discussed in [62], as well as the reward function designed to minimize frequency and tie-line power deviations (with penalties for suboptimal actions) in [63], introduces another layer of vulnerability. Poor tuning of these parameters could impair the learning process, resulting in suboptimal outcomes and reduced system stability.

2.4. Converter driven stability (CDS)

The power system inertia plays a vital role in maintaining frequency stability, as it helps the system resist frequency fluctuations following disturbances. However, the ongoing shift toward renewable energy sources, particularly converter-based resources, has led to a significant reduction in system inertia, thereby posing new challenges to grid stability. Consequently, accurate monitoring and estimation of system inertia have become essential for developing effective control strategies and ensuring reliable grid operations. To achieve this aim, research on the applications of DLTs for the forecasting and control of inertia in converter/ inverter-based power systems has been on the rise in recent years. In line with this, the existing literature can be broadly categorized into two main areas: deep reinforcement learning-based control applications and deep learning-based prediction applications. Fig. 7 depicts the core concepts driving this research.

Firstly, in terms of **deep reinforcement learning-based control applications**, several studies have proposed innovative approaches to address inertia-related challenges. For instance, considering the impact of impedance on the dynamics of power systems, Eskandar et al. [16] proposed an impedance emulation technique in a grid-forming inverter (GFI) to synthesize inertia in low X/R autonomous microgrids (MGs). This approach, referred to as DRL-GFI, effectively modifies the bandwidth of the inverter voltage controller to regulate inverter output power and, consequently, control frequency fluctuations. Similarly, Egbomwan et al. [67] introduced a twin delayed deep deterministic policy gradient RL technique for virtual inertia control, aimed at improving frequency instability in low inertia grid-connected and islanded MGs composed of inverter-based distributed generators. In addition, Yang et al. [65] devised a MADRL control strategy to optimally coordinate inertia and droop parameters of multiple energy storage systems (ESs) with virtual synchronous generators (VSGs) to dampen power oscillations arising from mismatched parameters of parallel coupled VSGs in power grid.

Table 7
The critical information based deep learning techniques for power system frequency stability.

Ref.No	Method	Model input features	Model output	Training strategies	Methods comparison, performance meterics and range of accuracy	Advantages of proposed method
[41]	CNN-LSTM	P_{ei} from t_0 to $3f_t$; ΔP_i from t_0 to $3f_t$; V_j from t_0 to $3f_t$; V_{gj} from t_0 to $3f_t$; Load P_i from t_0 to $3f_t$	Dynamic frequency re-sponse; optimal load shedding	- Measure topology data - High-D electric distance matric reduction to 2-D - Construct 3-D tensors - Convert into 4-D tensors - Transfer learning process	1. CNN-LSTM | **RMSE: 0.0042 - 0.0057** 2. ANN | RMSE: 0.0173 - 0.01923. 3. SVM | 0.0155 - 0.0181 4. EL | RMSE: 0.0159 - 0.0188	Enables proactive frequency forecasting to prevent system collapse.
[60]	MBEFC(1D-CNN, GRU)	G_{EP}; G_{MP}; L_P; V_A; V_ϕ	Frequency Response Safety Levels (FRSL); Frequency Response Safety Times (FRST).	- Data feature max-min normalization - Multi-branch learning - Ensemble connection - Multi-task training - Dropout - Channel attention block	1. Proposed | **ACC (FRSL): 98.79 % - 99.2 %: RMSE ($FRST_1$): 0.009777 - 0.01360** 2. DT | ACC (FRSL): 94.98 % - 96.97 %; RMSE ($FRST_1$): 0.01432 - 0.03182 3. RF | ACC (FRSL): 96.10 % - 97.85 %; RMSE ($FRST_1$): 0.01408 - 0.02868 4. SVM | ACC (FRSL): 94.89 % - 97.07 %; RMSE ($FRST_1$): 0.01513 - 0.03328 5. ANN | ACC (FRSL): 96.60 % - 97.34 %; RMSE ($FRST_1$): 0.01385 - 0.02703 6. SDAE | ACC (FRSL): 97.29 % - 97.48 %; RMSE ($FRST_1$): 0.01309 - 0.01621 7. 2D-CNN | ACC (FRSL): 95.60 % - 96.84 %; RMSE ($FRST_1$): 0.01313 - 0.01481	Delivers accurate frequency stability analysis with ro-bustness to measurement noise and data loss.
[44]	LSTM-RNN	frequency fluctu-ations	Power fluctu-ations	- Data collection & normalization - Modelinitialization & training - De-normalization &output	1. Curve fitting | MAE: 2.791; MAPE: 138.22 % 2. BPNN | MAE: 0.039 - 0.218; MAPE: 25.38 %- 34.98 % 3. RBFNN | MAE: 0.038 - 0.076: MAPE: 105.72 % - 158.72 % 4. GRU | MAE:0.15–0.162; MAPE:15.39 %-18.39 % 5. LSTM |**MAE:0.002–0.014; MAPE:6.23 %-24.27 %**	Enables accurate detection of power fluctuations for improved frequency control and storage placement.
[61]	MADRL	Local area control error (ACE) signal	Frequency deviations; unscheduled tie-line power flows	- Optimize agents param-eters using Multi-agent DDPG - Initialize training with PID database - Training DNN with SDAE refind with single-agent DDPG - Decentralized implementation	1. Fine-tuned PID | MA-ACE: 0.037 % - 0.042 % 2. Deep Q-learning | MA-ACE: 0.061 % - 0.093 % 3. Single-agent DDPG | MA-ACE: 0.028 % - 0.038 % 4. Proposed method | MA-ACE: 0.023 % - 0.029 %	Delivers superior perfor-mance in cooperative load frequency control (LFC).
[62]	Critic-attention-MARL	ACE signal	Regulated power generation	- Critical network with attention mechanism; - Share parameters - Reward function with cooperative mechanism - Experience replay	1. PID | Mean ACE: 0.00102 % - 0.026 % 2.DQN | Mean ACE: 0.00186 % - 0.0351 % 3. MADDPG | Mean ACE: 0.00122 % - 0.0255 % 4. proposed | **Mean ACE: 0.00067 % - 0.0186 %**	Attention mechanism en-ables selective focus and faster training.
[63]	TD3	Local area control error signal	- Minimize frequency deviations and tie-line power fluctuations - Optimal PID gain values	- delayed update for actors - dual critics and actors - Inclusion of clipped noise to control actions - Experience replay memory	1. Proposed TD3 | **IAE: 0.0003047 - 0.0005287** 2. DDPG | IAE: 0.0006915 - 0.001206 3. PSO | IAE: 0.0005977 - 0.001099 4. GA | IAE: 0.0009787 - 0.001398	Reduces settling time and frequency/powerdeviations by up to 60 %.

Table 8
Intrinsic advantages of proposed DLTs from frequency stability studies.

Ref. No	Method	Intrinsic advantage
[60]	MBEFC (1D-CNN, GRU)	Improves diverse feature learning pathways by integrating multiple branches (1D-CNNs and GRUs) through ensemble, enhancing robustness and generalization.
[61,65,66]	MADRL	Centralized training with decentralized execution enables coordinated policy optimization using only local observations at runtime.
[62]	Critic attention-MARL	Accelerates convergence by enabling agents to focus selectively on the most relevant state information, reducing the dimensionality of learned interactions.
[63]	TD3	Reduces overestimation bias in Q-value estimation through dual critics and delayed policy updates, leading to more stable and reliable policy learning in continuous control tasks.
		(Check for the intrinsic advantages of Ref. [41,44] methods in Table 3)

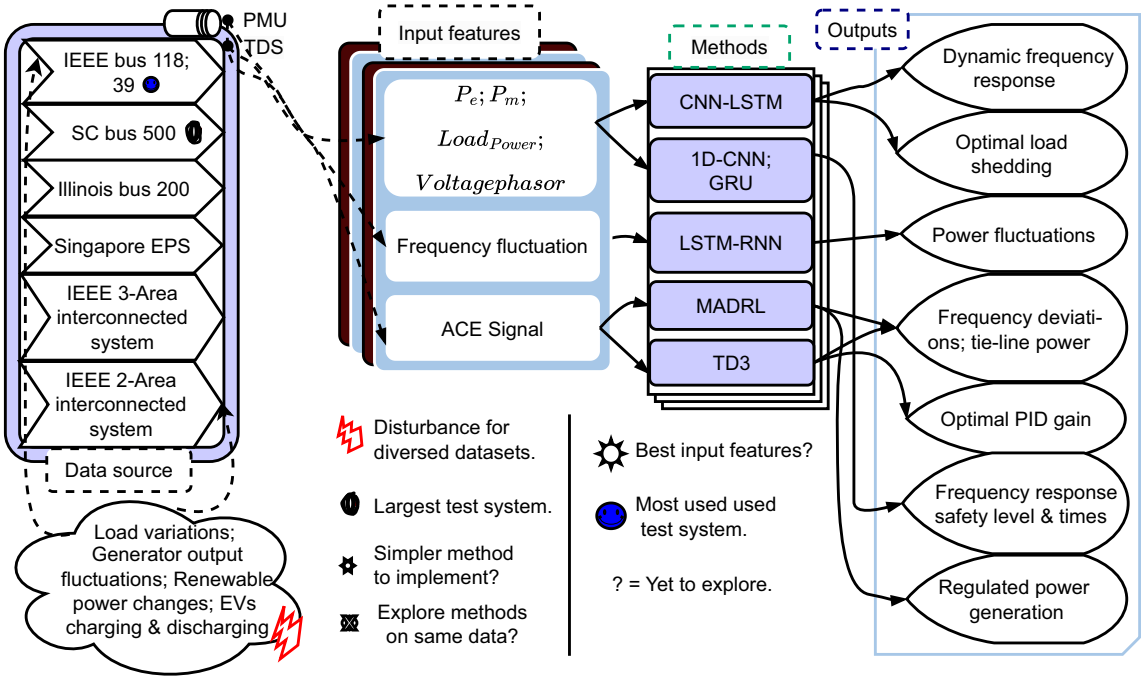


Fig. 6. Insightful framework for robust development of DLT's model for frequency stability and for further research consideration.

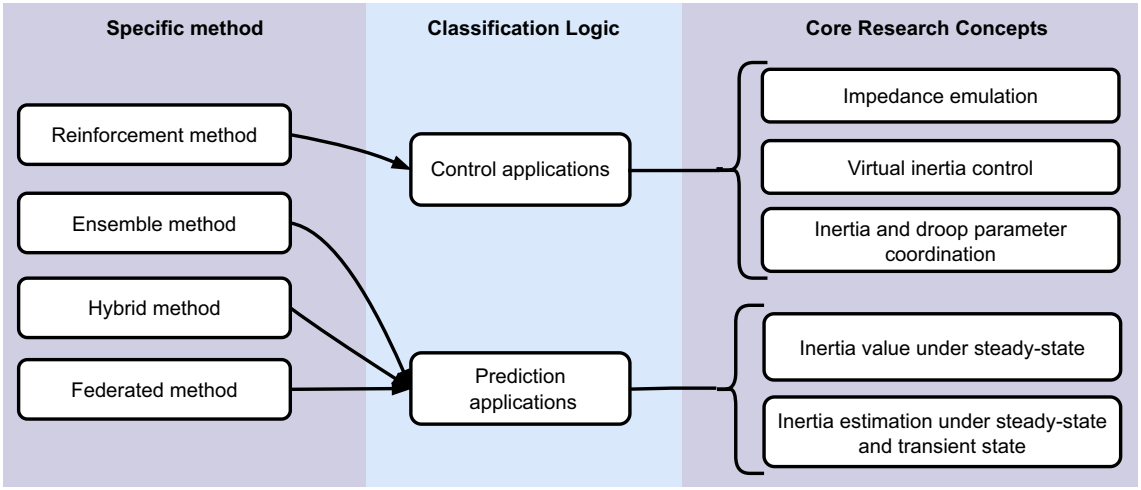


Fig. 7. The application of DLTs for power systems converter driven stability: core concepts central to the current research.

On the other hand, with respect to **deep learning-based prediction applications**, several researchers have focused on accurately forecasting system inertia under different operating conditions. For example, Muhammed et al. [7] proposed a novel dual-stage hybrid deep learning ensemble model incorporating long short-term memory (LSTM) networks, temporal convolutional networks (TCN), and convolutional LSTM (ConvLSTM) to accurately predict system inertia. This ensemble model is designed to address the challenges of fluctuating inertia resulting from high levels of renewable energy sources (RES) integration. In addition, Paidi et al. [45] utilized feed-forward neural networks (FFNNs) and multilayer perceptrons (MLPs) to forecast system inertia under normal operating conditions of power systems with high-power wind plant penetration. Similarly, Schmitt et al. [47] devised a perceptron neural network (PNN) to estimate inertia of buses, regions, and the overall system under normal conditions. In another study, Tuo et al. [68] employed long-recurrent convolutional neural networks (LRCNs) and graphical convolutional neural networks (GCNs) to estimate system inertia under steady-state operating conditions. Moreover, Kisinga et al. [46] developed an artificial neural network (ANN) model to estimate power system inertia under disturbance and steady-state conditions. Furthermore, Poudyal et al. [32] devised federated learning (FL)-based convolutional neural network architectures enabling decentralized estimation of area inertia through client-server network models. The critical information on the prediction and control of converter-driven stability in the current literature based on DLTs is presented in Table 9. Table 10 summarizes the intrinsic advantages of the proposed DLTs from converter-driven stability studies, and Fig. 8 groups existing concepts based on input characteristics and output configurations.

2.4.1. Challenges in current DLTs designs for CDS

Although the current design of DLTs for power system converter-driven stability forecasting and control demonstrates noteworthy advancements, it is not without limitations.

Table 9
The critical information based deep learning techniques for power system converter-driven stability.

Ref. No	Method	Model input features	Model output	Training strategies	Methods comparison, performance meterics and range of accuracy	Advantages of proposed method
[16]	DRL (actor-critic: two CNN)	Scalogram of instantaneous Active power	Optimal impedance shaping; gain matrix calculation	- Actor-critic (DDPG); - Continuous wavelet transform to transform observation - Reward function formulation - Learning updates - Implement actor-critic networks - setting target networks	1. DRL-GFI | **stabilizes the system** 2. Droop-controlled PI-based GFI | does not stabilize the system	Allows low-order control with quantified impedance–inertia link and smoother frequency response.
[67]	TD3-RL (actor-two critic)	Frequency deviation	Change in energy storage Active power	- Actor-critic method - Clipped double Q-learning - Delay actor network update than critic network - Target policy Smoothing - Learning parameter selection - Reward function - Twin delayed-DDPG (TD3)	1. Proposed | **MAE: 0.0106 - 0.0440**; **MAPE: 0.0186 - 0.0734** **RMSE: 0.0133 - 0.0619** 2. DDPG | MAE: 0.0204 - 0.0408; MAPE: 0.0340 - 0.0761; RMSE: 0.0269 - 0.0527 3. PI method | MAE: 0.0176 - 0.1244; MAPE: 0.0553 - 0.1544; RMSE: 0.0440 - 0.1720	Outperforms PI and DDPG by avoiding Q-value overestimation and accounting for inverter dynamics.
[65]	MADRL Markov game	VSG data: frequency; RoCoF	Inertia & droop parameters	- Soft-actor-critic (SAC) networks - Formulate problem as a multi-agent Markov game - Entropy-Augmented Reward - Actor-Critic Networks - Replay Buffers - Distributed training architecture	1. Central DRL 2. Conventional adaptive inertial control	Improves power oscillation damping over conventional adaptive inertia control.
[7]	Deep ensemble learning	P_m; P_e; RoCoF; P_{pv}	Aggregated Inertia Levels; System Inertia	- Data Normalization - Handling Missing Values - Time Delay handling - ADAM optimizer - Dropout & EarlyStopping - Grid search - Ensemble Model	1. **Proposed | RMSE: 0.0274** 2. LSTM-MLP | RMSE: 0.0351 3.TCN | RMSE: 0.0821 4. GRU | 0.1166 5. ConvLSTM | RMSE: 0.0718 6. TCN-LSTM | RMSE: 0.0508 7. LSTM | 0.0629 8. BiLSTM | RMSE: 0.0792 9. MLP | RMSE: 0.1482	Robust and generalizable under noisy or incomplete data.
[45]	FFNN; MLP	P_{SGs}; P_{IBRS}; induction(IM) load	System inertia constants	- Feature selection using Pearson Correlation Coefficient - Levenberg-Marquardt backpropagation - Hessian approximation for weight regularization	|	Incorporates demand-side inertia from induction motors.
[32]	FL with CNN	Frequency, tie-line power, RoCof	Area inertia constant	- Local Training & Decentralization - Client & server method - Federated Averaging-Offline Training and Online Updates	1. FL + CNN | RMSE: 0.534 - 0.5372 2. FL + MLP | RMSE: 0.372 - 0.4484	Enhances security via CNN feature weight sharing.
[68]	LRCN, GCN	Frequency deviation, RoCoF,voltage	System inertia constant	- Wrapper feature selection - Activation: ReLU - Dynamic learning rate - Backpropagation	1. LRCN | ACC: 97.34 %; Coefficient of determinations (R^2): 0.9725 2. GCN | ACC: 98.15 %; R^2: 0.9826 3. CNN | ACC: 95.18 %; R^2: 0.9369 4. DNN | ACC: 93.45 %; R^2: 0.9224	Improves inertia estimation using graph data, outperforming DNN and CNN.
[46]	Forward ANN	RoCoF, power variation	Inertia constant	- Sigmoid activation in hidden layer - ReLU activation in output layer - Levenberg-Marquardt backpropagation algorithm	|	Enables online inertia estimation.
[47]	PNN	Modes dampings frequency	Bus, regional, system inertia	- Earlystooping - Bagging	|	Improves the accuracy of regional and system inertia estimation.

Table 10
Intrinsic advantages of proposed DLTs from converter driven stability studies.

Ref. No	Method	Intrinsic advantage
[16]	DRL (actor-critic: two CNN)	Using CNNs as both actor and critic in DRL model facilitates effective feature extraction from high-dimensional temporal inputs.
[67]	TD3-RL (actor-two critic)	Reduced overestimation bias in value function approximation through clipped double Q-learning with two critic networks, a known limitation in standard actor-critic methods like DDPG.
[7,69,70]	Deep ensemble method	Improved predictive generalization through variance reduction by averaging outputs from diverse hybrid models.
[32]	FL with CNN	Preserves data privacy by transmitting only model weights instead of raw data, thereby reducing cybersecurity risks.
[68]	LRCN	Combines spatial feature extraction via CNN with temporal sequence modeling via LSTM, enabling robust representation of time-series data with spatial dependencies.
[68]	GCN	captures topological relationships and inter-node dependencies in non-Euclidean structured data through graph-based data passing.
		(Check for the intrinsic advantages of Ref. [65] method in Table 8 and [45–47] methods in Table 3)

First, a key shortcoming of ensemble methods proposed in [7] lies in the assumption of uniform weighting among individual models. Hence, these methods ignore adaptive weighting schemes that could account for varying input data characteristics or operating conditions. As a result, performance may become suboptimal, particularly in scenarios where certain models are better suited to specific data patterns or disturbances. Similarly, the reward function design in [16], which guides the deep reinforcement learning (DRL) agent by incorporating essential stability metrics such as frequency nadir and settling time, neglects other critical factors such as energy efficiency, power quality, and network interactions. Thus, this limited focus could lead to optimization decisions that, while improving inertia synthesis and frequency stability, might compromise broader operational objectives and grid performance. Moreover, training algorithms such as TD3 [67] introduces further challenges. Specifically, the approach requires meticulous hyperparameter tuning, making it computationally demanding and potentially unsuitable for real-time applications. In addition, the reliance on partially observable multi-agent reinforcement learning frameworks [65] introduces issues of non-stationarity and partial observability. In such environments, dynamic interactions among agents invalidate stationarity assumptions, which may result in unstable convergence or suboptimal policies.

Furthermore, the feature selection process based on Pearson Correlation Coefficient (PCC) analysis in [45] may inadequately capture non-linear relationships in scenarios involving complex interactions, such as those with renewable energy sources. Therefore, this limitation can reduce the accuracy and robustness of forecasting tools. Communication challenges also impact frameworks like federated learning [32]. In practice, unreliable communication links, bandwidth constraints, and synchronization difficulties can undermine weight aggregation and weaken overall robustness of the system. In addition, scalability concerns emerge in graph convolutional network models [68], particularly as power system size grows to thousands of nodes; without more advanced graph processing techniques, computational complexity becomes prohibitive. Finally, the dependence on steady-state conditions in [47] restricts the applicability of the proposed method under dynamic operating environments. Therefore, these techniques may fail to capture rapid system changes or transient events. Together, all of these limitations collectively underscore the need for further refinement and innovation in DLTs-based solutions for forecasting and control of converter-driven stability.

2.5. Voltage stability

Voltage stability refers to a power system's capability to sustain consistent voltage levels at all buses when subjected to a disturbance. To facilitate rapid estimation of voltage stability, without resorting to computationally intensive nonlinear iterative methods, DLTs are increasingly being developed and applied to address several voltage stability challenges in this domain. Consequently, this section categorizes the existing literature into two principal areas: deep reinforcement learning-based control applications and deep learning-based prediction applications. Fig. 9 encapsulates the core concepts central to the present research.

On the one hand, in terms of **deep reinforcement learning-based control applications**, several researchers have proposed innovative

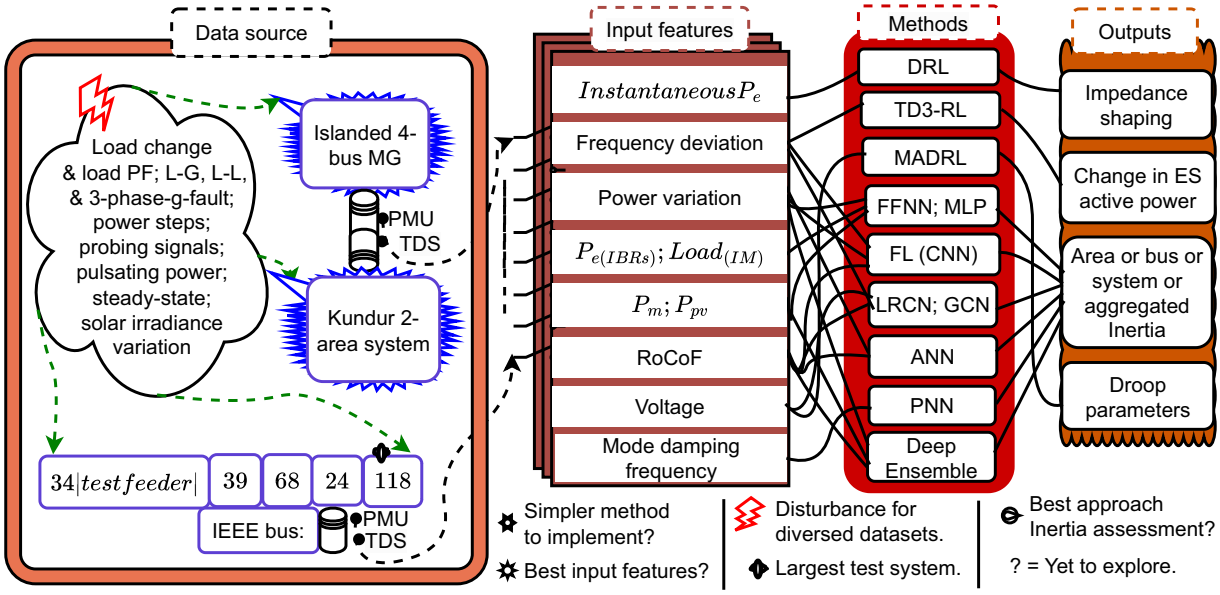


Fig. 8. Insightful framework for robust development of DLT's model for CDS and for further research consideration.

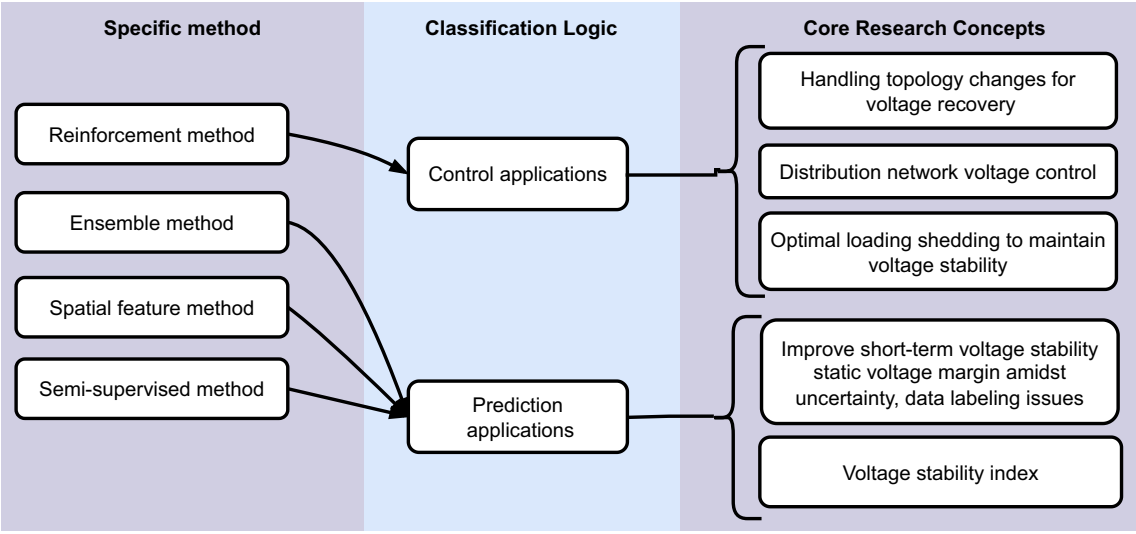


Fig. 9. The application of DLTs for power systems voltage stability: core concepts central to the current research.

methods to control voltage stability issues under different operating conditions. For instance, Hossain et al. [72] utilized GCN to assist deep reinforcement learning (DDQN) in capturing topology changes and spatial features, thereby improving the control of fault-induced delayed voltage recovery (FIDVR). Beyond FIDVR issue, Zhang et al. [73] developed a multi-agent GCN-attention scheme for emergency voltage control to restore voltage evolution above transient voltage recovery threshold. In this approach, GCN-based agents extract graph-structured voltage features to improve accuracy, while the attention mechanism enables cooperative learning among agents in dynamic grid environments. Similarly, Reference [74] proposed an automatic voltage control scheme for power grids that uses TL and MADRL to remediate the degradation of model-free methods amid changes in the topology of the power network. Furthermore, to optimize distribution networks (DNs), Cao et al. [75] presented a hybrid method combining the soft actor-critic DRL algorithm with a power flow surrogate model to generate reward signals. This approach delivers robust voltage control even under model inaccuracies and abnormal measurements. In a related effort, Cao et al. [66] developed MADRL to derive coordinated PV inverter control strategies for voltage regulation in DN with a high penetration of PV sources. In another contribution, Wang et al. [76] proposed a model-free DRL strategy for scheduling energy storage systems (ESS), aiming to reduce voltage deviation costs in unbalanced low-voltage DNs. This approach can approximately solve the multistage quadratic stochastic problem of ESS voltage regulation. Similarly, Mu et al. [77] emphasized that maintaining voltage within a safe operating range is critical amidst intermittent renewable energy (RE); hence, they designed a decentralized partially observable Markov decision process (POMDP) and a multi-agent GCN-based RL (MAGRL) algorithm to effectively manage voltage control in active DNs. In addition, Huang et al. [81] developed a derivative-free DRL algorithm (P-ARS), specifically customized for controlling voltage stability via load shedding. Likewise, Reference [39] developed a deep feedback learning machine (D-FLM) for adaptive and economical undervoltage load shedding (U-VLS), capable of forecasting voltage violations and adjusting its strategy using stepwise constraint relaxation and incremental learning for enhanced dependability. Building on this, Zhang et al. [82] utilized a deep Q-network (DQN) to tackle U-VLS through optimal load-shedding approach to maintain voltage stability. In this method, convLSTM network is used to learn the evolving features of short-term voltage instability, and a new reward function is introduced that includes voltage recovery consideration and DQN initialization. In contrast to load-focused strategies, Reference [83] proposed a deep deterministic policy gradient (DDPG)-based DRL method to continuously regulate the reference voltage of generator excitation systems during disturbances, thereby supporting voltage stability through device-level control.

On the other hand, concerning **deep learning-based prediction applications**, good progress has also been made. For example, to address the challenges posed by changing system topology, sample labeling limitations, and small datasets in short-term voltage stability (STVS) evaluation, Li et al. [71] employed a deep transfer learning approach. Their model leverages real-time PMU data, integrates temporal ensembling (TE) for sample labeling, and uses least squares generative adversarial networks (LS-GANs) for data augmentation to improve adaptability. Similarly, [78] customized CNN for STVS evaluation using network topology information and data from distribution network reconfiguration (DNR). Additionally, Reference [80] employed constraint-partitioning k-means (COP_k-means), a semi-supervised clustering algorithm, to label STVS samples and subsequently trained an LSTM model for real-time monitoring of system stability. In a related study, Reference [59] incorporated stochastic DNR data into a CNN to forecast the voltage stability index of radial networks under uncertain generation and load conditions. In contrast, Su and Lai [69] addressed the complexity of evaluating static voltage stability under uncertain conditions such as renewable variability, variable power demand, and aging transmission networks by proposing a deep ensemble learning framework. This method integrates enhanced deep quantile regression (e-DQR) models with an improved Choquet fuzzy integral (e-CHI) to estimate probabilistic static voltage stability margins (VSM). Similarly, Wu et al. [70] developed a deep ensemble model based on an extreme learning machine (ELM). The model allows for simultaneous point and probabilistic forecasts, enabling a bidirectional evaluation of VSM even during fluctuations of PV power. To tackle the challenge of limited labeled data for SVS monitoring, Reference [79] employed a disagreement-based deep neural network (DNN) method and proposed a TL strategy to adjust the model using a minimal set of labeled samples reflecting new operational scenarios. Lastly, Seyedi et al. [84] utilized a bidirectional LSTM (BLSTM) model trained on short-time mean squared voltage deviations from interlinking converters in islanded low-voltage hybrid AC/DC microgrids to classify post-fault voltage stability status based on device output voltage. Tables 11 and 12 present the critical information on the prediction and control of voltage dynamics in the current literature based on DLTs' methods. Table 13 summarizes the intrinsic advantages of the proposed DLTs from voltage dynamics studies,

Table 11
The critical information based deep learning techniques for power system voltage stability.

Ref. No	Method	Model input features	Model output	Training strategies	Methods comparison, performance meterics and range of accuracy	Advantages of proposed method
[71]	TL-Trans-former	Normalized volt-age(V), active power(P), reactive power(Q)	Stability status	- TE for sample labeling - LS-GANs for data augmentation	1. Proposed | F1-score: 0.9784 2. LSTM | F1-score: 0.9236 3. DT | F1-score: 0.9013 4. SVM | F1-score: 0.8137	Improves accuracy (20 %) using TL and self-attention; handles small data and labeling issues.
[72]	GCN-DDQN	Load information	improved voltage recovery	|	1. Proposed | parameters: 5768 2. FC-DDQN | 134,088	Enhances voltage recov-ery with 30 % less load shedding vs. FCN-DDQN.
[73]	GCN-MARL (actor-attention -critic)	Bus voltage deviations	Adjusted load shedding	- Central offline training - Off-policy SAC - Markov game - Q-learning objective function	1. MADQN | NEC: 954; Average re-ward after 800 episodes (ARAE): 1063.1 2. MASAC | NEC: 765; ARAE: 4161.5 3. MAAC | NEC: 547; ARA-800-E: 4392.2 4. Proposed | **NEC: 112; ARAE: 4769.7**	Improves EVC by en-hancing adaptability and ensuring system stability.
[74]	MADDPG + Transfer learning	Voltage magni-tudes VR, TC, CR	Q-power dis-tribution; minimized voltage deviation	- Markov game decision	1. MADDPG | CSR: 94.6 % 2. MAPPO | CSR: 93.9 %; 3. QMIX | CSR: 92.5 % 4. DDPG | CSR: 89.6 %,; 5. PPO | CSR: 87.9 %; 6. SAC | CSR: 90.2 %	Enables effective control within a restricted zone.
[75]	DRL + Power flow surrogate model	Branch P and Q of 1 % Gaussian noise	Controlled node voltage	- Soft-actor-critic - Physics-Informed GGAT - Physics-Informed DAE	1. Original (SAC = 0) | Ave: 1.76 %; Max: 8.02 %; Min: 6.96 % 2. Partial | Ave: 0.41 % - 1.23 % Max: 3.04 % - 7.35 %; Min: 3.16 % - 5.58 % 3. BP-SAC | Ave: 0.46 % - 0.81 %; Max: 3.28 % - 8.10 %; Min: 3.15 % - 4.06 % 4. GCN-SAC | Ave: 0.43 % - 0.70 %; Max: 2.42 % - 6.42 %; Min: 3.47 % - 4.01 5. Proposed | **Ave: 0.40 % - 0.50 %;** **Max: 2.54 % - 4.38 %; Min: 3.33 % - 3.59 %** 6. Global | Ave: 0.40 %; Max: 2.96 %; Min: 3.30 %	Robust to bad data; re-duces DN parameter reliance; stabilizes voltage from partial observations.
[66]	MA-DRL	$P_{L,t}$, $Q_{L,t}$, $P_{PV,t}$	$Q_{PV,t}$; ad-justed voltage deviation	- Training local policy networks and cen-tral critic networks in opposition to each other - MADDPG + attention model	1. 1. Original | Ave: 1.68 % - 1.89 %; CT: − 2. Droop | Ave: 0.46 % - 1.26 %; CT: 0.0002s 3. MADDPG | Ave: 0.16 % - 0.58 %; CT: 0.0012s 4. **Proposed | Ave: 0.12 % - 0.53 %; CT: 0.0012s** 5. Centralized | Ave: 0.04 % - 0.42 %; CT: 0.67s - 1.32s	Achieves coordinated PV inverter control using local data, minimizing com-munication infrastructure costs.
[76]	MADDPG al-gorithm + attention	Active power; states of charge (SOC); time stamp	compensated voltage	- MDP decision - Adam optimizer	|	Dynamically reduces daily voltage regulation cost via near-optimal strategy.
[77]	MAGRL + Barrier function	load P & load Q; Photovoltaic (PV) P & Q; Voltage phasors	Minimized volt-age deviation & reduced network loses	- Actor-critic framework; - Addition of RNN to actor network - Exponential barrier function added reward function - Experience replay - soft update strategy	1. Proposed | **CR: 0.977 - 0.992; PL: 0.049 - 1.153**; **QL: 0.178 - 0.792; VD: 0.013 - 0.013** 2. MADDPG-Exp | CR: 0.928 - 0.959; PL: 0.056 - 1.597; QL: 0.131 - 0.767; VD: 0.014 - 0.017 3. MADDPG-Bowl | CR: 0.898 - 0.892; PL: 0.068 - 1.836; QL: 0.137 - 0.602; VD: 0.013 - 0.021 4. Droop Control | CR: 1.000 - 1.000; PL: 0.082 - 1.519; QL: 0.177 - 0.476; VD: 0.009 - 0.011	Provides dynamic voltage regulation to maintain ±5 % stability amid renewable variability.
[59]	CNN	P & Q injection at bus; bus $\|V\|$ & <V; branch switch status	Steady-state stability; Short-term volt-age stability (RVSI)	- SDNR information - Branch reduction SBR algorithm	|	Speeds up SDNR solution vs. conventional MINLP solvers.
[78]	CNN + SDR	Active power; reactive power	RVSI	- K-medoid, t-SNE of clustering	|	Identifies optimal topolo-gies to improve STVS via DNR.
[69]	Deep ensemble learning (e-DQR, e-CHI)	Load P, load Q, generation P	VSMs	|	1. PQR | p-loss: 12.52 - 14.07; CRPS: 11.13 - 12.98; 2. LQR | p-loss: 11,75 - 13.68; CRPS: 11.08 - 12.47 3. GP | p-loss: 11.07 - 12.93; CRPS: 10.11 - 11.58 4. SVQR | p-loss: 10.42 - 11.78; CRPS: 9.09 - 10.85 5. MOR | p-loss: 9.77 - 11.33; CRPS: 8.61 −10.41 6. QRNN | p-loss: 9.11 - 10.66; CRPS: 8.03 - 9.66 7. FPSQR | p-loss: 9.01 - 10.25; CRPS: 7.87 - 9.38 8. HQGRU | p-loss: 8.75 - 10.03; CRPS: 7.76 - 9.02 9. **Proposed | p-loss: 5.26 - 6.62; CRPS: 4.65 - 5.48**	Provides accurate and adaptable estimation under system changes.
[70]	Deep ensemble of ELMs	Load P & Q; bus voltage; load/PVs directions; solar irradiance; tem-perature; system topology	VSM^+ & inter-val VSM^- & interval	- Optimization blend - K-means	1. **Proposed | MAPE for VSM^+: 0.62 % - 3.07 %**; **MAPE for VSM^-: 1.4 % - 2.65 %** 2. MLP | MAPE for VSM^+: 4.32 % - 5.64 %; MAPE for VSM^-: 4.88 % - 5.53 % 3. RF | MAPE for VSM^+: 4.16 % - 5.79 %; MAPE for VSM^-: 4.58 % - 5.21 % 4. SVM | MAPE for VSM^+: 3.91 % - 4.95 %; MAPE for VSM^-: 4.07 % - 5.23 %	Offers robust VSM gener-alization amid growing PV integration.

Table 12
The critical information based deep learning techniques for power system voltage stability.

Ref. No	Method	Model input features	Model output	Training strategies	Methods comparison, performance meterics and range of accuracy	Advantages of proposed method
[79]	Disagreement DNN	P_L, Q_L, P_G, Q_G, $\|V\|$, $\angle V$	Voltage stability label	- Transfer learning scheme	1. SVC | ACC: 46.36 % - 86.51 % 2. LinearSVC | ACC: 74.62 % - 91.67 % 3. Radial basis SVC (NuSVC) | ACC: 77.62 % - 93.43 % 4. RF | ACC: 64.71 % - 95.33 % 5. AdaBoostSAMME.R (AB_R) | ACC: 60.22 % - 97.19 % 6. AdaBoostSAMME.D (AB_D) | ACC: 61.61 % - 96.67 % 7. DNN | ACC: 71.70 % - 96.56 % 8. **Proposed: ACC: 88.94 % - 99.00 %**	Achieves high accuracy with minimal labeled data, reducing labeling costs.
[80]	COP_k-means + LSTM	Voltage, P, Q	Stability status	- Adam optimizer - Dropout method -	1. Proposed | ACC: 95.08 % - 98.36 %; F1-score: 0.9479 −0.9825; AUC: 0.9855 - 0.9963 2. DT | ACC: 92.21 % - 93.44 %; F1-score: 0.9183 0.9313; AUC: 0.9392 - 0.9482 3. SVM | ACC: 87.70 % - 87.70 %; F1-score: 0.8646 0.8646; AUC: 0.9509 - 0.9781	Enables accurate and timely STVS assessment.
[81]	P-ARS + LSTM	Bus voltage magnitudes; % of load at controlled buses	Minimized total load shedding & voltage violations	- Simultaneous perturbation stochastic approximation - LSTM	1. P-ARS | ACT: 0.72 s 2. MPC | ACT: 63.31	Delivers fast convergence, robust learning, and strong scalability.
[39]	D-FLM (LSTM, FC network)	Bus V, P, Q load shed action	Next interval Voltage violation index (VVI)	|	1. Proposed | $R_{fal/\%}$: 0; $\bar{p}_{shed}$/MW: 760.37; $V\bar{V}I$/pu: 0.0287; $\bar{V}_{dev}$/pu: 0.0892 2. DRL | $R_{fal/\%}$: 0; $\bar{p}_{shed}$/MW: 751.42; $V\bar{V}I$/pu: 0.0290; $\bar{V}_{dev}$/pu: 0.0899 3. ELS | $R_{fal/\%}$: 9.93; $\bar{p}_{shed}$/MW: 704.62; $V\bar{V}I$/pu: 0.0374; $\bar{V}_{dev}$/pu: 0.1217 4. USLS | $R_{fal/\%}$: 0; $\bar{p}_{shed}$/MW: 930.35; $V\bar{V}I$/pu: 0.0326; $\bar{V}_{dev}$/pu: 0.1061	Directly evaluates future voltage impacts of UVLS measures.
[82]	DQN, ConvLSTM	P, V	Optimal load shedding	- Modified reward function	1. **Proposed | load shedding/p.u (LS/p.u):0.752; Voltage recovery index/p.u (VRI/p.u): 0.522** 2. IMI-based scheme | LS/p.u: 1.078; VRI/p.u: 0.465 3. Practical approach | LS/p.u: 1.772; VRI/p.u: 0.503	Reduces load shedding by over 58 % and improves voltage recovery under uncertainty.
[83]	DRL (DDPG)	Bus voltage levels; ΔV_t	Adjusted excitation voltage response	- DDPG algorithm - Markov decision process	|	Effectively restores bus voltages post-disturbance compared to conventional methods.
[84]	BLSTM	V_{DC}, V_{AC}	Voltage stability status	- Adam optimizer - Regularization coefficient	1. Proposed | lowest FPR 2. LSTM | 3. DT | 4. SVM |	Delivers superior performance with low false positive rate.

and Fig. 10 groups existing concepts based on input characteristics and output configurations.

3. Power systems cyber security enhancement using deep learning-based methods

3.1. Bad data/ adversarial examples/ false data

The increasing reliance on digital systems and communication networks in modern power grids has introduced new vulnerabilities, particularly to false data injection (FDI), adversarial examples, and bad data. These threats can compromise the accuracy of monitoring systems, disrupt grid operations, and, in the worst case, jeopardize overall grid stability. In response, DLTs offer promising solutions for detecting and preventing such threats through robust anomaly detection and control modeling frameworks. Therefore, this section divides the existing literature into two primary categories: deep learning-based detection approaches and deep reinforcement learning-based control strategies. Fig. 11 illustrates the core concepts underlying the present research.

On the one hand, in the domain of **deep learning-based detection applications**, innovative methods have been developed for detecting bad and false data injection issues. For example, to address the challenge of performing accurate prefault dynamic security assessments using incomplete PMU data, [85] proposed a generative adversarial network (GAN) composed of two deep learning networks. This method can generate missing data irrespective of PMU observability or system topologies. In contrast, to improve the detection of stealthy FDI attacks (S_FDIA), Yin et al. [40] developed deep PowerFDNet to identify S_FDIA from spatiotemporal measurements in AC transmission systems. Similarly, Ahmadzadeh et al. [42] proposed a deep learning-based detection strategy using CNN to identify S_FDIA within the voltage control system of a PV-integrated power distribution system (P_DS). In addition, Moayyed et al. [43] presented a unique method for detecting FDI attacks in

Table 13
Intrinsic advantages of proposed DLTs from power system voltage stability studies.

Ref. No	Method	Intrinsic advantage
[71]	TL-Transformer	Enables efficient learning of long-range temporal dependencies and cross-domain generalization through self-attention and pretraining adaptability.
[72]	GCN-DDQN	Enables topology-aware feature extraction by embedding spatial correlations of nodal data, allowing superior generalization and policy transfer across varying graph structures.
[73]	GCN-MARL (actor-attention-critic)	Facilitates efficient cooperative learning by combining graph-based topological encoding and attention-driven inter-agent communication.
[74]	MADDPG + Transfer learning	Improves learning efficiency by enabling policy reuse across environments without retraining, even under structural or parametric changes.
[75]	DRL + Power flow surrogate model	Reduces dependence on accurate physical system models by learning reward signals from data-driven surrogate models.
[66]	MADDPG algorithm + attention	Improved scalability through selective relevance modeling, allowing each agent's critic to focus on the most impactful interactions among agents.
[77]	MAGRL +Barrier function	Enhanced representation of agent interactions and system topology through graph based model integration improves policy learning stability and scalability in decentralized multi-agent environments.
[79]	Disagreement DNN	Reduces reliance on manual labeling by leveraging model disagreement to pseudo-label unlabeled data, enabling efficient semi-supervised learning.
[80]	COP k-means + LSTM	Improved label reliability through semi-supervised constraint-based clustering enhances LSTM model training by reducing label noise without requiring extensive manual annotation.
[81]	P-ARS + LSTM	It enables robust training without back-propagation, avoiding exploding gradients in non-smooth environments, due to its simultaneous perturbation stochastic approximation approach.
[82]	DQN, ConvLSTM	Enables automatic extraction of translation-invariant spatial–temporal features, enhancing decision-making in settings with dynamic and structured data.
[83]	DRL (DDPG)	DDPG can efficiently learn optimal control policies in environments with continuous action spaces by combining actor-critic architecture and deterministic policy gradients.
[84]	BLSTM	It can capture both past and future temporal dependencies in sequential data, enabling accurate pattern learning.
		(Check for the intrinsic advantages of Ref. [39] method in Table 3, [59] method in Table 6, [66] method in Table 8 and [69,70] methods in Table 10)

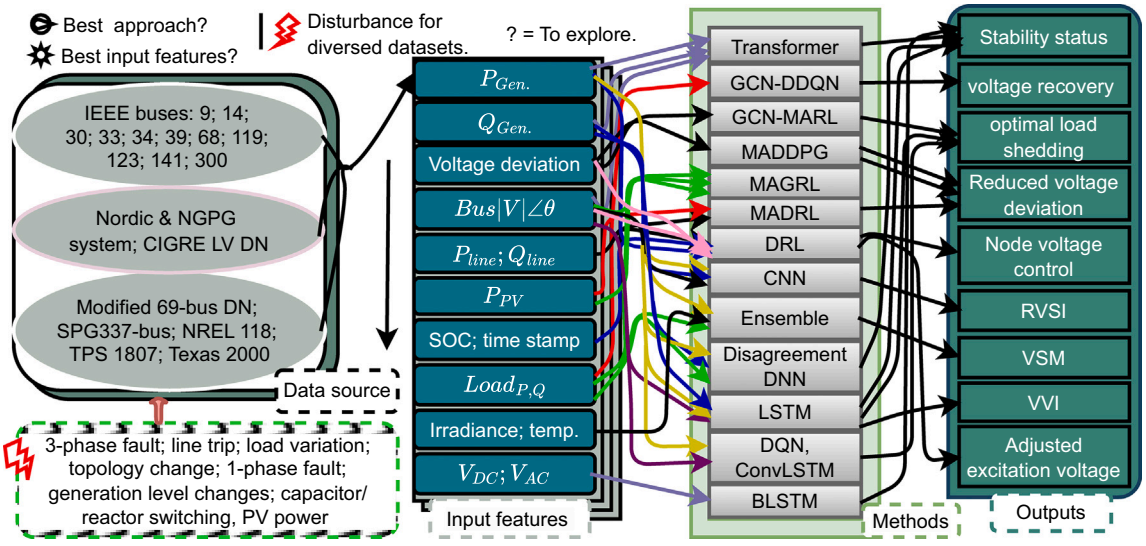


Fig. 10. Insightful framework for robust development of DLT's model for voltage stability and for further research consideration.

power grids. The approach involves converting multivariate system state time-series data into 2D colored images, which are then analyzed using a deep CNN classifier to enable human operators to visually identify anomalies and trends. Likewise, Wang et al. [91] developed a CNN-enhanced location detection scheme that integrates with a standard bad data detector (BDD) to identify the exact locations of FDI attacks in smart grids. Moreover, to safeguard privacy in FDI detection processes, Li et al. [87] devised a hybrid technique that incorporates transformer models, federated learning (FL), and the Paillier cryptosystem (PC). In this architecture, transformers deployed at edge nodes utilize multi-head self-attention to capture inter-node data relationships, while FL allows for collaborative model training without centralized data storage, and PC ensures encrypted communication to protect against data breaches. In another study, Feng et al. [90] proposed an unsupervised scheme combining dual graph-convolutional autoencoders (DAE) and GAN to detect FDI attacks. The method uses graph convolution to improve the representation of high-dimensional non-Euclidean grid data and dynamic thresholds for adaptive anomaly detection. Conversely, Liu et al. [86] focused on attack development by designing a data injection method that compromises the critical clearing time (CCT) estimates generated by the ELM strategy in transient stability assessment. In this method, a deep convolutional neural network (DCNN)-based generator—comprising a node selector, filter, encoder, and decoder—is trained via GAN to produce stealthy and effective attack vectors by identifying vulnerable nodes, refining attack strategies, extracting relevant features, and generating disturbances.

On the other hand, research on **deep reinforcement learning-based applications** remains relatively limited. For instance, to prevent FDI attacks targeting distance relays, capable of mimicking faults and triggering unnecessary trips, [88] proposed a multi-agent distributed deep

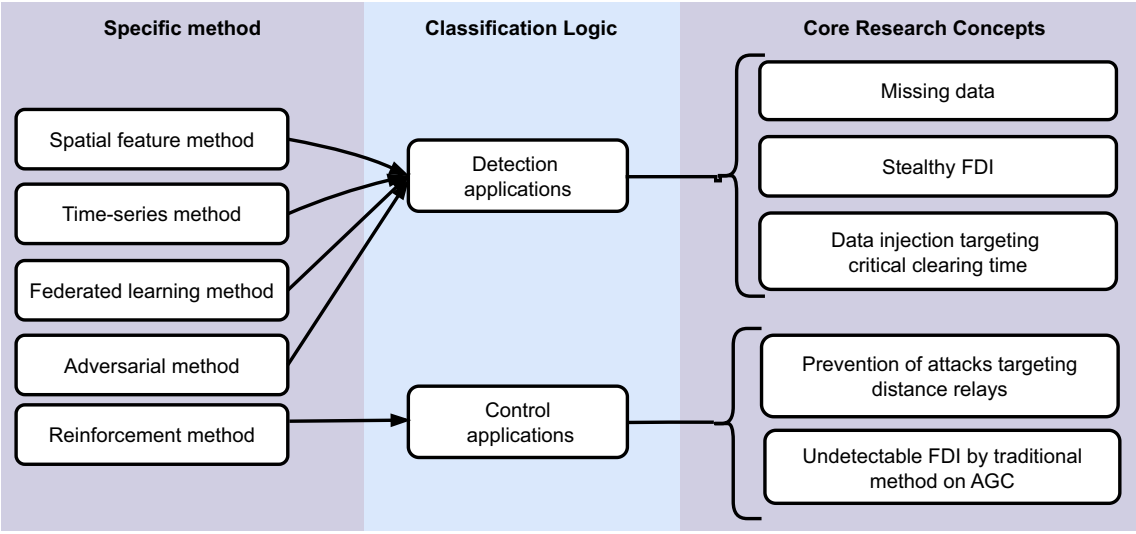


Fig. 11. The application of DLTs for bad data/ Adversarial examples/ false data: core concepts central to the current research.

learning (MADDL) framework. In this approach, each distance relay functions as an agent within a multi-agent system, sharing voltage and current data with neighboring relays through a graph-theoretic structure. Cyberattack detection is then carried out by deep neural networks assigned to each agent, which collectively analyze local and neighboring data inputs. Furthermore, [89] investigated the vulnerability of AGC systems to stealthy FDI attacks that are undetectable by conventional methods. The authors employed an RL approach to devise attack policies capable of bypassing traditional defenses such as the area control error (ACE) and the unknown input observer (UIO). By framing the attack scenario as a multi-objective partially observable Markov decision process (MO_POMDP), they utilized the proximal policy optimization (PPO) algorithm to learn and optimize these covert strategies. Table 14 presents critical information on the detection of FDI attacks in the current literature based on DLT methods and Table 15 summarizes the intrinsic advantages of the proposed DLTs from FDI studies. Fig. 12 groups existing concepts based on input characteristics and output configurations.

3.1.1. Challenges in current DLTs designs for bad/ false data detection

The current design of DLTs for detecting bad data or FDI attacks shows promise but is constrained by certain limitations. One major challenge concerns the handling of missing data. Although existing methods are adaptable to diverse PMU placement strategies, they fail to address systemic or correlated failures across multiple PMU. This shortcoming can compromise the effectiveness of GAN, which rely on consistent data patterns for accurate detection [85]. In addition, these approaches typically assume that measurement noise follows a Gaussian distribution and remains minimal—less than 1 % for voltage magnitude and 2 % for power flow. This assumption oversimplifies real-world scenarios, where both noise and data loss are often more severe, thereby affecting detection accuracy [40]. Compounding this issue is the challenge in differentiating between anomalies caused by stealthy FDI attacks and those arising from legitimate system events, such as equipment malfunctions or line outages [42]. Moreover, some detection methods require extensive prior knowledge of attack targets, including access to input-output datasets and communication protocols, an assumption that is often unrealistic, particularly in highly secure or adversarial environments [86,89]. Furthermore, many approaches focus narrowly on S_FDIA, while neglecting other forms of attacks that may occur simultaneously or sequentially, limiting their real-world applicability [87,90]. Another vulnerability arises from the reliance on neighboring agents or relays for accurate data sharing. Inaccuracies in these components can propagate system-wide errors, especially in the absence of mechanisms to validate the integrity of shared data. As a result, these systems remain susceptible to coordinated attacks targeting multiple relays simultaneously [88]. Finally, the common practice of employing traditional bad data detectors as a pre-filter introduces additional risk. Sophisticated FDI attacks specifically designed to evade such mechanisms can bypass initial defenses, rendering the overall system ineffective [91].

3.2. Cyber physical intrusion

The increasing integration of digital systems and Internet of Things (IoT) devices into power grids has significantly expanded the attack surface for cyber-physical intrusions. These intrusions, which exploit vulnerabilities in cyber physical components of the grid [100], can result in severe disruptions such as cascading failures, equipment damage, and loss of operational control. In response to these threats, DLTs are emerging as powerful tools for detecting cyber-physical intrusions by analyzing complex data patterns and enabling real-time decision-making. Accordingly, this section discusses **deep learning-based detection approaches**, which dominate current literature on cyber-physical security of smart grid. Fig. 13 presents the core concepts that form the basis of current research.

To address the cybersecurity challenges associated with cyber-physical components in smart grid, which, while enabling bidirectional power flow, also increase vulnerability to cyber threats, Kaur et al. [93] proposed a CNN-Bayesian approach. This method integrates Bayesian techniques to distinguish between cyber-physical intrusions and normal events, while using CNN layers to handle high-dimensional input data effectively. Similarly, Keshk et al. [95] developed a privacy-preserving framework to secure smart power networks against data privacy violations and cyber-attacks. Their solution comprises two main modules: a two-level privacy module and an anomaly detection module. The former employs a proof-of-work blockchain (PoWB) to maintain data integrity against poisoning attacks and incorporates a variational autoencoder (VAE) to encode data and prevent inference attacks. The latter module utilizes an LSTM-based deep learning scheme to validate the outputs of the privacy module. On the other hand, Behdadnia et al. [92] demonstrated an attack strategy that exploits the learning capabilities of CNN-LSTM to detect vulnerable states of power system operations from encrypted PMU data. This approach can enable system operators to implement countermeasures to reduce the success rate of attackers in using encrypted data analysis schemes to execute timely denial-of-service

Table 14
The critical information based deep learning techniques for bad data/ adversarial examples/ false data problems.

Ref. No	Method	Model input features	Model output	Training strategies	Methods comparison, performance meterics and range of accuracy	Advantages of proposed method
[85]	GAN	Incomplete PMU data	Complete PMU data	- GAN model training - Adam optimizer - Training hybrid ensemble learning model (ELM, RVFL network) - Use randomized learning algorithm	1. Proposed | ADAI: 99.04 % - 99.40 % 2. REL | ADAI: 97.96 % - 98.48 % 3. DTWS Method | ADAI: 80.81 % - 83.28 % 4. Feature Estimation | ADAI: 96.12 % - 96.99 %	Accurately reconstructs missing PMU data.
[40]	PowerFDNet (CNN, LSTM)	Bus P, Q, V line P, Q, I	S_FDIA label non-S_FDIA label	- Spatiotemporal Modeling - Residual connections in CNN; Adam optimizer - ReduceLROnPlateau scheduler	1. M-I | F1-score: 94.140 % - 95.085 % 2. M-II | F1-score: 95.359 % - 96.250 % 3. PowerFDNet | F1-score: 99.267 % - 99.668 %	Achieves higher S_FDIA detection accuracy.
[42]	CNN	Seasons, weekdays, load conditions, PV generation bus voltage	Normal & attack conditions	Data preparation; feature standardization and normalization; model design and training; Dropout and early stopping	1. Proposed | F1-score: 96.26 - 97.36 2. MLP | F1-score: 94.87 - 96.82 3. SVM | F1-score: 94.59 - 96.92	Demonstrates superior performance over existing methods.
[86]	DCNN + GAN	Node V, power angle (δ), P, Q	Adversarial examples of input data	- Adversarial training of GAN	1. Proposed | ATSA: 0.26ms; DR: 5.3 % 2. DE algorithm | ATSA: 145.8 s; DR: 32.7 %	Generates bad data to support attack-resilience methods.
[87]	Transformer + FL + PC	$\|V\|$ & $\angle$V, bus & line P deviation	False normal data	Secure FL; Initialization clients & parameters; Client local training; Communication encryption; Aggregation of model Weights Updating local models	1. Proposed | Round 6 F1-score: 0.9344 - 1.000 2. LSTM | Round 6 F1-score: 0.6210 - 1.000 3. CNN | Round 6 F1-score: 0.5371 - 1.000	Preserves data privacy and reduces communication rounds.
[88]	MADDL + RNN	Voltage, current	Detected normal, fault, attack data	Agent training data; Training of DNN for each agent; Evaluation & testing; Distributed implementation	|	Achieves over 99.8 % detection accuracy for faults and attacks.
[43]	Deep CNN	$\|V\|$, $\angle$V 2D image	Clean and attack data detection	- Use of ReLU, Dropout, time-series to images Adam optimizer	1. Deep CNN | ACC: 93.34 % - 100 %; F1-score: 93.00 % - 100 % 2. SVM | ACC: 92.36 % - 97.23 %; F1-score: 92.00 % - 97.00 %	Provides high visualization accuracy of FDIAs across networks.
[89]	RL (PPO)	Frequency (F) & P deviations	Detection of stealthy attacks	Problem formulation as a MO-POMDP; use PPO for training RL; flexible reward function; tanh activation; stochastic gradient descent (SGD) to optimize actor-critic NNs	|	Reveals stealthy RL-based attacks that disrupt system frequency.
[90]	DAE + GAN	Bus & branch power	Normal & attack	Used historical data & anomaly scores to train the model	1. Proposed | acc: 99.11 % - 99.90 % 2. A3D | acc: 95.74 % - 99.58 % 3. LSTM-AE | 95.54 % - 99.14 %	Demonstrates robustness under noisy conditions.
[91]	CNN + BDD	Buses & lines power flow, $\|V\|$ residuals from BDD	Locational detection of F_DIAs	Adam: a patience of 5, Cross-entropy loss, Mini-batches, Cross-validation	1. MLP | F1-score: 97.05 - 1.35 2. CNN | F1-score: 99.40 - 99.75	Accurately locates F_DIAs within the smart grid.

Table 15
Intrinsic advantages of proposed dlts from bad data/ adversarial examples/ false data studies.

Ref. No	Method	Intrinsic advantage
[85]	GAN	Ability to generate data samples that match the distribution of training data without requiring an explicit parametric model.
[86]	DCNN + GAN	Ability to generate highly similar adversarial samples with minimal perturbations, enabling real-time generation while evading detection mechanisms.
[87]	Transformer + FL + PC	Enables privacy-preserving, distributed feature extraction with encrypted inter-node collaboration.
[88]	MADDL + RNN	Allows scalable, parallel processing of time-series data through decentralized RNN agents, reducing computational costs.
[89]	RL(PPO)	PPO enables stable and efficient policy learning in high-dimensional, continuous action spaces, making it well-suited for optimizing sequential decision problems under uncertainty.
[90]	DAE + GAN	Improves feature representation and robustness by jointly optimizing reconstruction and latent consistency via adversarial training and dual encoders, independent of application context.
[91]	CNN + BDD	Model-free architecture requiring no prior statistical assumptions, enabling generalization across systems without altering existing BDD frameworks.
		(Check for the intrinsic advantages of Ref. [40,42,43] methods in Table 3)

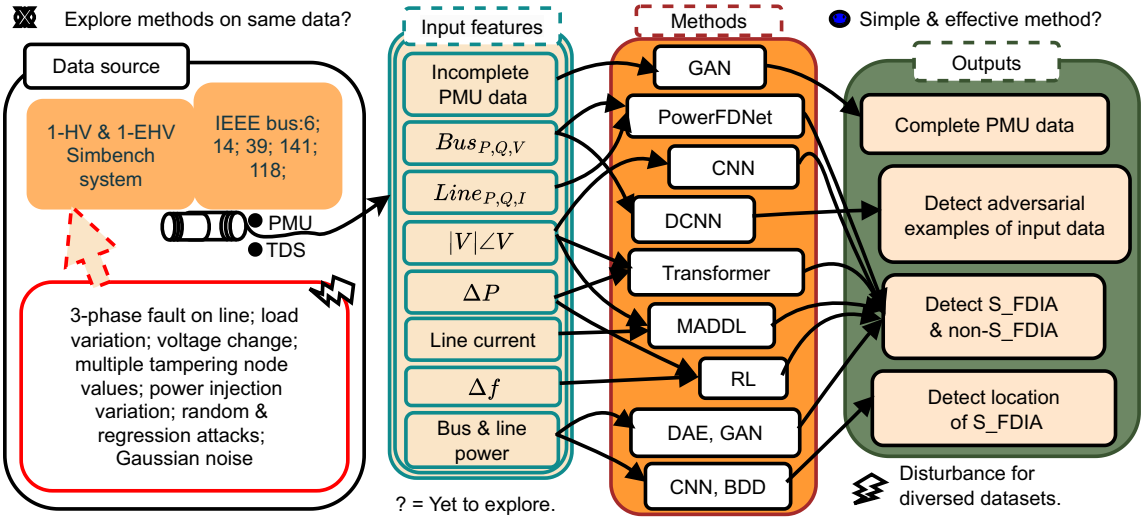


Fig. 12. Insightful framework for robust development of DLT's model for FDI and for further research consideration.

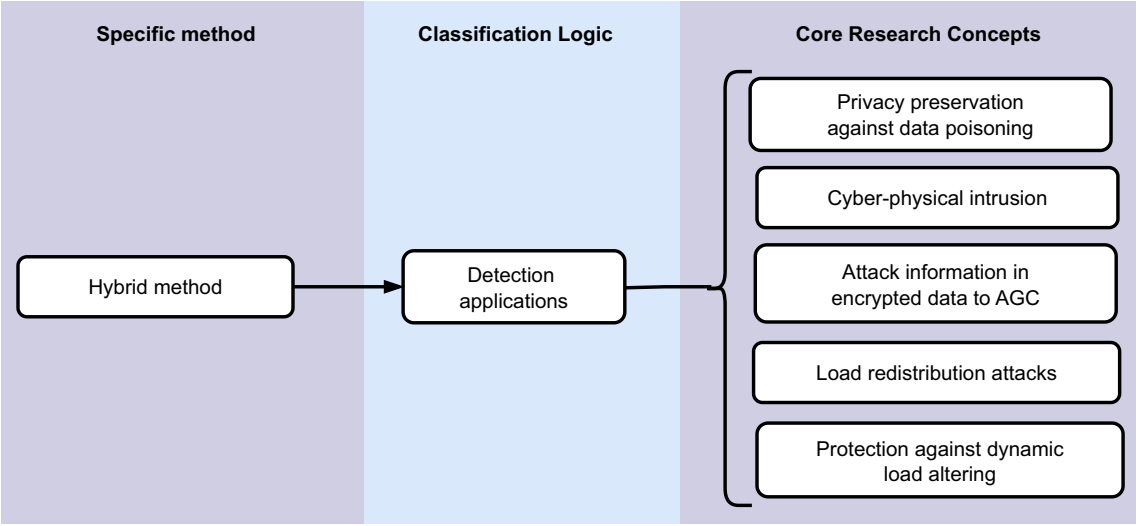


Fig. 13. The application of dlts for cyber physical intrusion: core concepts central to the current research.

(DoS) attacks on automatic generation control systems. In a related study, Khaleghi et al. [96] introduced a deep learning-based framework for detecting load redistribution (LR) attacks: malicious modifications of load and power flow data that can trigger cascading failures and blackouts. Their approach models LR attacks and uses an RNN to analyze cyber load data from the energy management system. Furthermore, to defend against dynamic load altering (DLA) attacks, launched to manipulate high-wattage IoT devices to destabilize power systems, Jahangir et al. [97] proposed a 2D CNN classifier and reconstruction decoder (2DRCNN). This model is designed to detect and localize DLA attacks by analyzing generator bus data, specifically focusing on phase angle and frequency variations. Wang et al. [98] tackled cyber threats under different smart grid operational scenarios by proposing a two-stage sparse cyberattack model. Their defense mechanism, based on interval state estimation, uses deep stacked autoencoders (SAE) to improve electric load forecasting accuracy, thereby enhancing anomaly detection capabilities. In addition, Khaw et al. [99] presented a cyberattack detection system for transmission line protective relays in substations. Their model utilizes a 1-D convolutional autoencoder (1D-CAE) to identify anomalies in current and voltage measurements, which could indicate the onset of a cyberattack. Moreover, recognizing that many existing attack detection methods rely on anomalies in power system measurements, often detected late in the attack process, Presekal et al. [94] proposed an early-stage stealthy cyberattack identification and localization method. This approach integrates software-defined networking (SDN) with a graph-based model that captures communication-level anomalies. The authors employed a hybrid deep learning model that combines graph convolutional long short-term memory (GC_LSTM) with a deep CNN to perform time series classification for improved anomaly detection and enhanced power grid resilience. Table 16 presents the critical information on cyber-physical intrusion detection in the current literature based on DLTs. Table 17 summarizes the intrinsic advantages of the proposed DLTs from cyber-physical intrusion studies and Fig. 14 groups existing concepts based on input characteristics and output configurations.

3.2.1. *Challenges in current DLTs designs for cyber physical intrusion*

The current design of DLTs for cyber-physical intrusion detection demonstrates promising performance; however, there are limitations. For example, the approach proposed in [92] relies on detecting subtle statistical variations in encrypted PMU signals. However, this reliance raises concerns about robustness, as enhanced encryption protocols such as AES-CBC and AES-CTR are specifically designed to obscure

Table 16
The critical information based deep learning techniques for cyber physical intrusion.

Ref. No	Method	Model input features	Model output	Training strategies	Methods comparison, performance meterics and range of accuracy	Advantages of proposed method
[92]	CNN + LSTM	Encrypted IPsec-protected PMU data (frequency deviations)	Vulnerability status	Data preparation & labeling; PMU data encryption; feature engineering; input into a matrix form; training and validation; sliding time window	1. Proposed + AES-CBC encryption | Precision: 0.814 2. Proposed + AES-CTR encryption | Precision: 0.816	Effectively blocks malicious AGC data analysis.
[93]	CNN-Bayesian	Normal; attack data	No-event, neutral, attack status	Data preprocessing; feature extraction; Bayesian layer integration; log-likelihood loss using Adam optimizer.	1. CNN-Bayesian | F1-score: 77.84; AUC: 0.84 2. CNN | f1-score: 48.87; AUC: 0.71 3. LSTM | F1-score: 45.29; AUC: 0.73	Performs well with high-uncertainty datasets.
[94]	GC_LSTM + deep CNN + SDN	OT network traffic data	Normal; attack status	GC_LSTM training, deep CNN training, implement hybrid model, tuning with Bayesian	1. Proposed | AUC: 0.9243 - 0.984; ACC: 91.25 % −96.45 % 2. GC-LSTM + Resnet | AUC: 0.924 −0.974; ACC: 90.02 % - 96.42 % 3. GC-LSTM + Inception | AUC: 0.9489 - 0.976; ACC: 90.05 % - 95.01 %	Identifies active attack locations with >96 % accuracy.
[95]	PoWB + VAE + LSTM	Recorded data	Abnormal; normal status	Two-level privacy-preserving module (Blockchain, VAE); anomaly detection module	1. Proposed | DR: 96.27 % - 99.80 %; FPR: 0.01 % - 2.93 % 2. PPAD-CPS | DR: 93.82 % - 96.26 %; FPR: 3.52 % - 6.75 % 3. CART | DR: 93.45 % - 94.93 %; FPR: 4.61 % - 6.53 %	Outperforms existing methods in effectiveness.
[96]	LR model + RNN	Bus load vector; load ratio	Normal LR attack; secured state; cascading failure attack	Data generation; offline training; online detection	|	Achieves detection accuracy between 94.1 % and 99.8 %.
[97]	2DR_CNN	Buses phase angle, buses frequency fluctuation	DLA detection, attack location	Data labeling, offline training, online inference, data augmentation, noisy data handling, detection & reconstruction loss	1. 2DR_CNN | acc: 99.27 % - 99.66 % 2. 1D_CNN | acc: 96.54 % - 97.49 % 3. AE_DNN | acc: 96.11 % - 96.36 % 4. SVM | acc: 91.34 % - 95.13 %	Robust against noise and missing data.
[98]	Deep SAE	$\|V\|$, $\angle V$	Detecting & localizing load altering threat (LAA)	Layer pretraining, batch processing, gradient descent, error modeling	1. SAE | average MAPE (A_MAPE): 3.51 % 2. SVM | A_MAPE: 5.10 % 3. BP algorithm | A_MAPE: 5.94 %	Highly effective in detecting sparse cyberattacks.
[99]	1D-CAE	Lines currents,line voltages	Cyberattack detection	Early-stopping, Sliding window method, data standardization, Adam optimizer, grid search	Proposed | Recall: 87.4 % - 100 %	Effectively detects cyberattack-induced anomalies.

Table 17
Intrinsic advantages of proposed dlts from cyber physical intrusion studies.

Ref. No	Method	Intrinsic advantage
[93]	CNN-Bayesian	Quantifies model uncertainty via probabilistic weight distributions, facilitating more reliable and interpretable predictions.
[94]	GC LSTM + deep CNN + SDN	It excels in a simultaneous learning of spatial and temporal features in a graph-structured data, enhancing prediction robustness and anomaly detection accuracy even with minimal data variability.
[95]	PoWB + VAE + LSTM	Provides resilient data integrity and privacy transformation through chained verification and generative encoding, enhancing robustness against data leaks and tampering.
[96]	LR model + RNN	RNNs inherently model short-term sequential dependencies, making them ideal for tracking load fluctuations and distinguishing abnormal shifts indicative of attacks.
[97]	2DRCNN	Joint spatiotemporal feature extraction with noise-robust reconstruction to improve model accuracy and resilience to data imperfections.
[98]	Deep SAE	Superior capability to extract nonlinear features and invariant structures from input data through hierarchical layer-wise learning.
[99]	1D-CAE	Efficient parameter sharing reduces training complexity while preserving local temporal patterns in sequential data.

such patterns. Consequently, even minor protocol modifications, such as adjustments to IPsec/ESP, could effectively eliminate these variations, rendering the method ineffective. Similarly, although the Bayesian methods employed in [93] offer strong capabilities for handling uncertainty, their efficacy diminishes in deterministic or low-noise datasets. In such cases, these methods may introduce unnecessary complexity without delivering significant performance improvements over simpler deterministic models. In contrast, the communication-focused anomaly detection technique described in [94] does not consider physical layer attacks that do not manifest in network traffic anomalies. This oversight limits its applicability by excluding critical physical system measurements and overlooking IT-OT interdependencies. Furthermore, while blockchain integration in [95] leverages a lighter ePoW mechanism, it remains susceptible to 51 % attacks. In addition, its reliance on

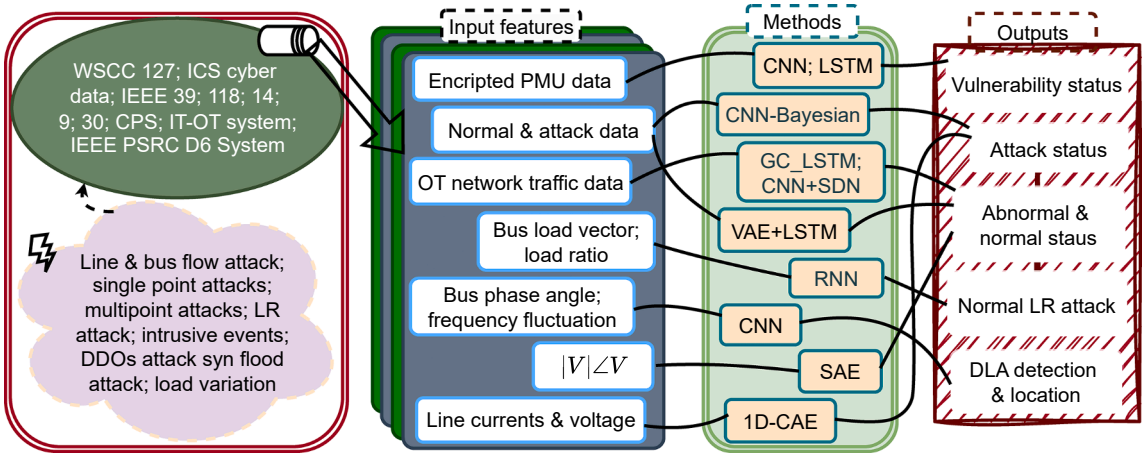


Fig. 14. Insightful framework for robust development of DLT's model for Cyber physical intrusion and for further research consideration.

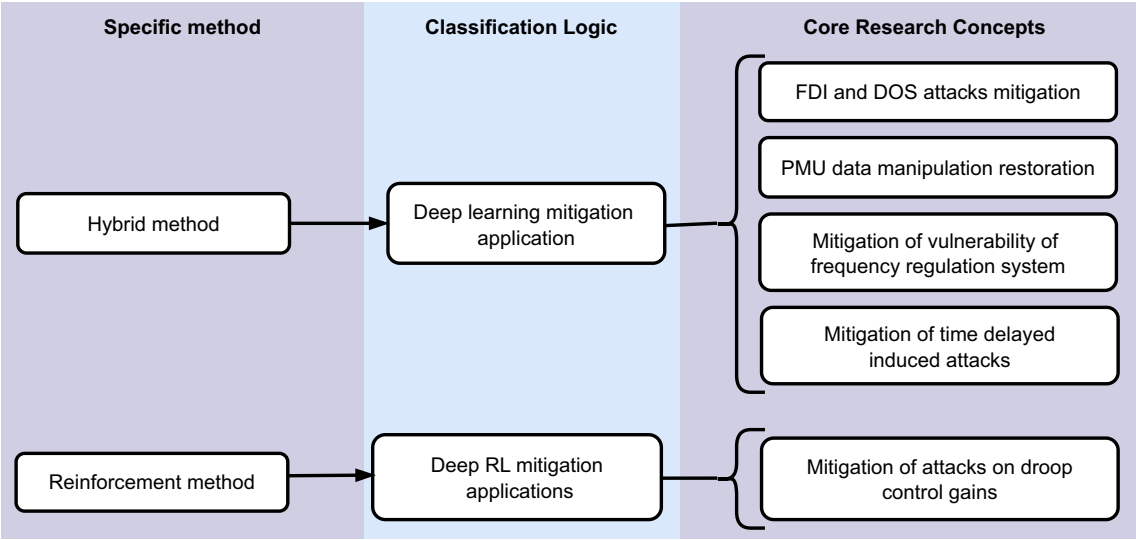


Fig. 15. The application of DLTs for Cyberattack Mitigation: core concepts central to the current research.

decentralized consensus mechanisms introduces the risk of delays or inconsistencies, particularly in unreliable or low-participation network environments. The method described in [96], which assumes highly resourceful adversaries executing worst-case LR attacks, may also prove inadequate against less sophisticated or stochastic threats that exploit different vulnerabilities. Likewise, the computational simplification achieved in [97] by relaxing the L0-norm to the L1-norm improves computational feasibility but may fail to capture attack configurations that deviate from these assumptions. Although [98] enhances robustness to noisy or incomplete PMU data through reconstruction networks, it may fall short in addressing correlated noise or advanced malicious data injections. Such attacks can compromise feature integrity, thereby undermining accurate detection and localization of load-altering threats. Finally, the system proposed in [99] is unable to detect fully synchronized replay attacks, wherein malicious data mimics legitimate fault conditions. This limitation allows such attacks to bypass differentiation mechanisms, posing a significant threat to system integrity.

3.3. Cyberattack mitigation

Cyberattack mitigation involves the deployment of proactive and reactive strategies to minimize the impact of attacks, restore normal operations, and improve system resilience. In response to the growing sophistication of cyber threats, DLTs have emerged as advanced mitigation tools capable of not only detecting but also actively mitigating malicious activities. Hence, this section categorizes existing literature into two principal groups: deep learning-based mitigation approaches and deep reinforcement learning-based mitigation strategies. Fig. 15 depicts the core concepts driving this research.

Firstly, in terms of **deep learning-based mitigation approaches**, recent studies have proposed innovative frameworks to strengthen system security against cyberattacks. For instance, Ahmed et al. [101] introduced robust deep learning architectures designed to safeguard AGC systems against stealthy FDI and DoS attacks. Their framework comprises one detection model and two mitigation models specifically tailored for stealthy and coordinated attacks. The detection model effectively identifies the type and location of attacks, distinguishing between normal operations, photovoltaic (PV) injections, and cyberattacks. Meanwhile, the mitigation components, the comprehensive mitigation model (CMM) and the attacked class-based mitigation model (ACM), restore corrupted measurements with minimal error. In a similar vein, Elimam et al. [19] proposed a deep learning-based cybersecurity scheme for detecting and recovering PMU data from manipulation attacks. This approach utilizes two deep learning models:

Table 18
The critical information based deep learning techniques for cyberattack mitigation.

Ref. No	Method	Model input features	Model output	Training strategies	Methods comparison, performance meterics and range of accuracy	Advantages of proposed method
[101]	Hybrid LSTM-Dense; CMM;ACM	Frequency Deviations; Tie-Line Power Flows	No attack; attack on area frequency; attack on tie-line power; multiple coordinated attacks; corrected frequency deviations; Restored tie-line power flows	Dataset generation; preprocessing; model training; Adam optimizer learning rate decay	Hybrid LSTM-Dense | acc: 98.20 %; ACM | RMSE: 0.003463; MAE: 0.001208 CMM | RMSE: 0.003218; MAE:0.000762	Demonstrates high robustness and precision in detecting and mitigating diverse and stealthy cyber-attacks.
[19]	ANN with DAE	Bus $\|V\|$, $\angle V$	Normal & attack status; mitigated attack	Feature engineering; training two models; variable learning rate; Dropout; Exponential linear unit (ELU) activation function	1. Proposed | ACC: 99.93 %; MSE: 3.23×10^{-3} 2. GRU | ACC: 97.42 %; MSE: 2.88×10^{-4} 3. LSTM | ACC: 81.88 %; MSE: 2.06×10^{-4}	Accurate attack detection with low signal restoration error.
[102]	ANFIS-based ADM + TI-LADRC	Frequency deviation	Attack, non-attack detected, Mitigated attacked ACE	Dataset preparation; data splitting; ANFIS Training in MATLAB	|	Highly resilient to FDI, DoS, and TD cyberattacks.
[103]	TD3	Voltage; frequency	Adjusted droop gains of IBRs	- Used MDP & AMDP task formulation; Training uses an adversarial interaction and TD3 Algorithm; ReLU & tanh activation	1. TD3 | 2. DDPG |	Enables effective attacks and robust defense strategies.
[104]	DRL + RLS + RBF	Electric price $\angle$V matrix	Adjusted price, power output, voltage	Distributed Learning, actor-critic approach, Recursive Learning,	|	Provides optimal recovery against F_DIA.

an attack detection ANN model to identify the presence and location of attacks, and a data recovery ANN based on a denoising autoencoder (DAE) to reconstruct compromised signals. In addition, in an effort to address the vulnerability of power grid frequency regulation systems, [102] developed a deep learning-based attack detection and mitigation system. This system integrates a dual-loop tilt integral-linear active disturbance rejection control (TI-LADRC) mechanism with a feed-forward structure. It is capable of detecting and mitigating various types of cyberattacks, including DoS, FDI, and time delay (TD) attacks.

On the other hand, in the domain of **deep reinforcement learning-based mitigation strategies**, Reference [103] investigated data-induced attacks targeting the droop control gains of inverter-based resources (IBRs) in microgrids and developed a robust defense method using adversarial deep reinforcement learning (ADRL). Both attack and defense scenarios were formulated as Markov decision processes (MDPs) and adversarial MDPs, respectively, and solved using the twin delayed deep deterministic policy gradient algorithm. In contrast, Zhang et al. [104] proposed a resilient defense scheme based on a distributed DRL framework to counter FDI attacks in microgrids. Their method employs recursive least squares (RLS) to evaluate the impact on supply security and voltage stability, and a radial basis function (RBF) neural network to model the relationship between electricity prices and load demand. A distributed actor-network is then used to optimize network weights, thereby formulating an optimal defensive strategy that not only enhances microgrid autonomy but also accelerates learning efficiency. Table 18 presents the critical information on cyberattack mitigation in the current literature based on DLTs methods. Table 19 summarizes the intrinsic advantages of the proposed DLTs from cyberattack mitigation studies and Fig. 16 groups existing concepts based on input characteristics and output configurations.

Table 19
Intrinsic Advantages of Proposed DLTs from cyberattack mitigation studies.

Ref. No	Method	Intrinsic advantage
[101]	Hybrid LSTM-Dense	Combines temporal pattern learning of LSTM with nonlinear feature abstraction of dense layers.
[19]	ANN with DAE	Ability to approximate complex, nonlinear mappings between corrupted and original data due to their universal function approximation property.
[102]	ANFIS-based ADM	Offers adaptive, data-driven anomaly detection by learning complex input–output mappings without relying on explicit physical models.
[103]	TD3	Check Table 10 for its intrinsic advantage.
[104]	Distributed DRL with distributed actor	Enhances learning efficiency through parallelized sub-agent coordination while preserving local autonomy, thereby reducing computational complexity compared to centralized approaches.

3.3.1. Challenges in current DLTs designs for cyberattack mitigation

The existing designs of DLTs for cyberattack mitigation demonstrate considerable capability in detecting and mitigating attacks; however, they are not without limitations. One notable challenge is the presence of signal reconstruction artifacts during the data recovery phase, as reported in [19]. These artifacts, particularly at the onset of attacks, can result in suboptimal decision-making in sensitive applications such as wide-area damping control, where accurate and smooth reconstruction is essential. Moreover, the mitigation approach proposed in [102], which involves switching ACE inputs from real-time to estimated data during attack detection, relies on secure communication within a local area network for frequency deviation measurements. Nevertheless, this method assumes that local measurements remain uncompromised, an

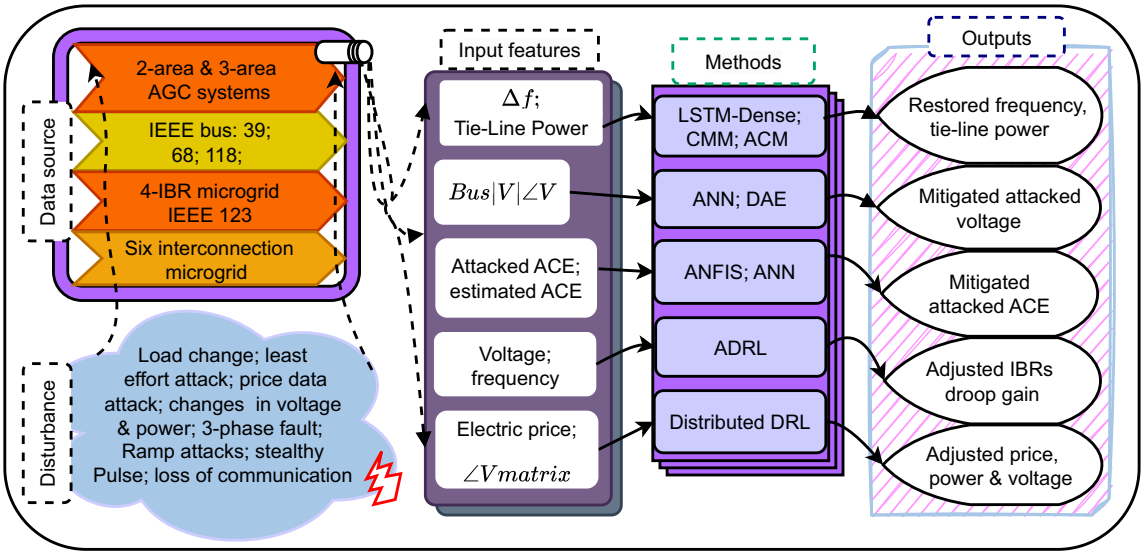


Fig. 16. Insightful framework for robust development of DLT's model for cyberattack mitigation and for further research consideration.

assumption that may not hold under advanced cyberattacks that exploit local vulnerabilities. In a similar vein, the strategy in [103] that targets droop control gain manipulation as the primary attack vector neglects other critical threats, including those to communication networks, data integrity, and control hardware. As a result, the mitigation strategy remains incomplete and potentially vulnerable. Furthermore, although the distributed DRL framework presented in [104] offers advantages such as reduced computational overhead and faster convergence, it is predicated on the assumption of robust, uninterrupted coordination and communication among microgrids. In practice, this assumption may not always be valid, especially in environments prone to communication delays, interruptions, or cyberattacks. Such issues can compromise system stability and ultimately diminish the effectiveness of the distributed framework.

4. Coupled phenomena (Cyberattacks to stability threats)

The preceding sections have reviewed applications of deep learning techniques (DLTs) to two domains traditionally studied in parallel: power system stability and cybersecurity. While separation is intended to have a clear focus on each phenomenon and to examine the algorithms and concepts used for system stability and cybersecurity individually, the discussion thus far may not by itself demonstrate why interdisciplinary research is necessary. Consequently, in this section, we articulate concrete reasons that should motivate a unified study of the phenomena, drawing upon evidence from the literature.

To begin with, cyber events have been shown to directly compromise the physical stability of power systems [105]. For example, false data injection attacks can systematically bypass conventional bad-data detection and corrupt state estimation, thereby misleading contingency analysis and dispatch decisions that underpin transient, voltage, and small-signal stability margins [106–108]. In [109], false data injection (FDI) attacks on storage-based transient stability control are examined. A parametric feedback linearization scheme is applied to detect and mitigate compromised rotor angle measurements, which, if left unaddressed, mislead controllers and trigger interarea oscillations and system-wide instability. Similarly, time-synchronization and GPS spoofing attacks on phasor measurement units alter synchrophasor streams used by wide-area monitoring and control, biasing dynamic stability assessments [110,111]. Collectively, these scenarios illustrate how falsified measurements can destabilize the system, since controllers like automatic voltage regulators, governors, and power system stabilizers operate based on incorrect setpoints [105].

Moreover, coordinated load-altering attacks launched through compromised IoT or EV charging devices have been demonstrated to excite electromechanical oscillations and threaten frequency stability [112, 113]. For instance, [114] shows that dynamic load-altering attacks can destabilize power systems by modifying active power via control signals. Such perturbations disrupt the generation–load balance, inducing small-signal instability, frequency oscillations, and rotor angle deviations that may ultimately trigger generator trips through protection mechanisms. Along similar lines, the authors in [115] introduce an FDI attack that manipulates load-bus power data to induce small-signal instabilities and raise operating costs, thereby showing that stability can be compromised either through load alterations or falsified control-center measurements. In addition, cyberattacks have been studied in the context of automatic generation control (AGC), where single- and coordinated stealthy false data injection attacks and denial-of-service attacks disrupt AGC functionality and trigger instability [101]. Likewise, attacks on data integrity can undermine coordinated voltage [104] or frequency control [102], with the potential to destabilize the system. Furthermore, historical incidents such as the 2015 and 2016 Ukraine grid cyberattacks underscore how adversarial access to control and monitoring infrastructure can push systems beyond secure operating regions and trigger large-scale instability [116]. In fact, a Proceedings of the IEEE article explicitly frames smart grid resilience as a cyber-physical challenge where control-loop vulnerabilities, communication delays, and physical dynamics cannot be decoupled [117]. Taken together, these findings establish that analyzing stability and cybersecurity separately obscures critical interdependencies that condition real-world reliability.

Equally important, deep learning itself is not merely a generic justification for interdisciplinary study but an enabling tool that plays a critical role in both domains. As reviewed in Sections 2 and 3, DLTs are already widely employed for stability assessment and control as well as for intrusion and anomaly detection and mitigation. However, DLTs' models are also vulnerable to adversarial perturbations and distribution shifts induced by cyber events, which render their performance unreliable unless explicitly designed for robustness [118]. This dual role means that DLT-enabled stability functions and DLT-enabled security functions create an inherent interdependence, since the robustness of each depends on the integrity of the other.

In summary, the integration of stability and cybersecurity is a matter of technical necessity. On the one hand, cyber compromises have already been shown to degrade or invalidate classical stability assessments. On the other hand, deep learning's growing role in both domains

introduces an additional layer of interdependence. Therefore, future research should move beyond isolated investigations and instead develop DLT-based methods that are designed to operate at the intersection of stability and security.

5. Future research directions

In order to fully realize the potential of deep learning techniques (DLTs) in modern power systems, future research should address the intertwined challenges of power system stability and cybersecurity in a unified manner. These domains are no longer independent—cyber-physical attacks can directly compromise system stability, while stability disturbances can mask or exacerbate cyber threats. As such, enhancing interpretability and scalability of DLTs should be pursued in an integrated framework that enables joint monitoring, diagnosis, and decision-making across both physical and cyber layers. One priority is improving the interpretability of deep learning models so that operators can understand and trust their outputs when assessing both stability margins and detecting cyber anomalies. For example, saliency-based interpretability methods can be applied to recurrent neural network variants used in transient stability assessment. They can be extended to flag cyber-induced measurement distortions, such as spoofed PMU signals, that might skew stability predictions. Similarly, attention-based Transformer architectures, when integrated with explanation techniques like SHAP or LIME, can simultaneously reveal the most influential stability-related features (e.g., bus voltages, rotor angles) and cyber indicators (e.g., abnormal SCADA command sequences, latency spikes), producing a unified situational awareness map. By embedding such cross-domain interpretability directly into decision-support dashboards, operators can trace critical actions—such as load shedding, islanding, or distributed energy resource (DER) dispatch—back to both the physical disturbances and cyber activities that influenced the AI model's recommendation. This integration can transform interpretability from a post-hoc analytical tool into a real-time operational enabler for managing the coupled threats.

In addition, scalability presents another critical challenge in these intertwined domains. Large-scale, geographically distributed power systems require real-time analysis of both stability and cybersecurity data streams, often under strict latency and bandwidth constraints. Centralized AI training and inference pipelines are inadequate for this task. Future architectures should adopt hybrid distributed learning approaches, such as federated learning combined with cloud–edge coordination, to scale AI models across multi-domain systems. In wide-area monitoring systems (WAMS), local substations can process PMU data for stability assessment and network traffic data for intrusion detection without transmitting raw inputs, thereby preserving both operational confidentiality and cyber log privacy. These local model updates can be aggregated in the cloud to maintain a synchronized global model, enabling cross-domain anomaly detection and stability prediction at scale. Edge-deployed distilled models can then make rapid decisions for tasks like voltage regulation, oscillation damping, adaptive protection, and cyberattack mitigation—ensuring responsiveness during both high-speed transients and coordinated adversarial events.

Equally important is the development of robust models capable of functioning under conditions of incomplete, noisy, or non-stationary data, challenges commonly encountered in real-world systems affected by sensor faults or cyber intrusions. Future research should incorporate uncertainty quantification techniques, such as Bayesian deep learning or ensemble methods, into forecasting and control models to generate probabilistic outputs and confidence bounds. In addition, the integrating noise-tolerant data imputation techniques, such as denoising autoencoders or generative models, can ensure continued operation of AI agents under partial observability, particularly in renewable forecasting or load estimation applications.

Furthermore, data privacy and security also demand increased attention, especially in decentralized architectures such as federated learning across distributed control centers and substations. Future research must prioritize the integration of privacy-preserving techniques, such as differential privacy and homomorphic encryption, into federated or edge-deployed DLTs. For instance, differential privacy can be implemented during gradient aggregation in federated load forecasting models to prevent inference of sensitive load profiles, while homomorphic encryption allows encrypted data processing in distributed attack detection systems without compromising confidentiality. Complementing these approaches, robust cybersecurity frameworks should incorporate adversarial machine learning techniques. Adversarially trained convolutional or graph neural networks can be evaluated against realistic attack scenarios for topology identification and false data injection detection. Future work should explore the use of generative adversarial networks and attack simulation environments to stress-test DLTs algorithms and develop certified defenses using robust optimization or randomized smoothing for dynamic security assessment and protection coordination. Moreover, embedding domain knowledge into DLTs' architectures has shown promise in improving both performance and interpretability. Physics-informed neural networks, which incorporate physical laws (e.g., swing equations, power flow constraints) into the learning process, should be investigated for dynamic stability modeling, particularly under high renewable penetration or contingency scenarios. Hybrid architectures that integrate data-driven models (e.g., LSTM, Transformer) with rule-based or model-predictive control strategies can be especially effective for DER coordination, where adaptability is crucial.

Additionally, real-time deployment demands that models operate reliably under diverse operating conditions and unforeseen scenarios. To address these challenges, a set of technically grounded and scenario-specific approaches can be considered. **Domain adaptation** techniques are essential for transferring models trained on high-fidelity transmission data to distribution-level systems with differing operational characteristics. For example, fault classification or voltage control models developed for one region can be rapidly adapted to new contexts without extensive retraining, thereby supporting scalable deployment. To further improve adaptability, ensemble learning strategies should be pursued. **Adaptive ensembles** that dynamically combine specialized models trained under various grid conditions, such as high DER penetration, islanded operation, or faulted states, can ensure sustained decision accuracy and responsiveness. **Hierarchical graph neural networks** (GNNs) also present a compelling avenue for modeling the multilevel structure of power systems. By capturing relationships across interdependent network layers, from feeders to substations and regional control centers, GNNs enable localized yet globally informed decision-making for tasks like fault localization and stability assessment. In parallel, **federated learning with asynchronous** update mechanisms offer a privacy-preserving, low-latency solution for distributed learning among diverse grid agents, including smart meters, microgrids, and substations. These mechanisms are particularly advantageous in environments with heterogeneous connectivity and processing capabilities, enabling real-time learning without centralized data collection. Furthermore, reinforcement learning models should incorporate **multiobjective reward formulations** to balance conflicting operational goals, such as voltage stability, frequency regulation, and cyber risk mitigation. This would enable distributed DLT-based control architectures to deliver not only high accuracy but also robust and holistic performance in real-time settings.

Another pressing challenge is the lack of standardized datasets and benchmarks, which hampers consistent progress and reproducibility across studies. Establishing open repositories and unified evaluation protocols can streamline research efforts and promote collaboration. In this context, interdisciplinary cooperation among AI researchers, power system engineers, and policymakers is vital to ensure that technological advancements align with practical requirements and regulatory standards. Looking ahead, emerging technologies such as quantum machine learning may offer breakthroughs in addressing complex optimization problems. Concurrently, advancing AI-driven solutions for microgrid

management can pave the way for autonomous and resilient energy networks. Hardware-in-the-loop testing should also be employed to rigorously evaluate the real-world performance of DLTs.

Finally, the growing prevalence of low-inertia, inverter-dominated power systems introduces unique challenges that traditional control mechanisms cannot adequately address. These systems, characterized by fast and nonlinear dynamics, necessitate advanced AI solutions, particularly reinforcement learning algorithms such as Proximal Policy Optimization and Multi-Agent Deep Reinforcement Learning, for coordinating synthetic inertia, adaptive damping, and frequency regulation across distributed assets. These models are well-suited for learning in environments with topology changes and uncertain conditions. Another under-researched yet increasingly relevant area is the emergence of interconverter oscillations in networks with high penetration of power electronic interfaces. These oscillations, often difficult to model and predict using traditional techniques, pose new challenges for maintaining small-signal and transient stability. Future research should apply graph-based deep learning architectures such as Graph Neural Networks and attention-enhanced Transformer models, which excel at capturing spatiotemporal dependencies. These models can be trained on high-resolution operational data to detect and mitigate such oscillatory interactions in a scalable and data-efficient manner.

6. Conclusion

The rapid evolution of power systems, driven by renewable energy integration, electrification, and digitalization, has introduced unprecedented challenges that demand innovative solutions. DLTs are at the forefront of addressing these challenges, offering solutions for enhanced stability and security. This paper has reviewed the state-of-the-art applications of DLTs across eight critical areas, highlighting the transformative potential of current DLTs' explorations and design limitations. Key insights include the ability of DLTs to handle the complexity of modern power systems, which can enable real-time monitoring, anomaly detection and mitigation, and predictive control. For example, well-trained deep learning models, such as stacked LSTM, CNN-LSTM, and Conv-LSTM, have shown significant improvements in LFOs forecasting by effectively capturing poorly damped oscillations in renewable-rich power systems, even under partial observability and missing data. Physics-informed deep learning and transfer learning hold promise for enhancing transient stability prediction, particularly for large rotor angle scenarios, even under highly variable conditions. Hybrid CNN-LSTM networks and RL strategies have demonstrated strong potential for frequency forecasting and control, enabling more accurate and efficient frequency regulation. Additionally, DRL, multi-agent systems, and ensemble learning are emerging as leading methods for estimating power system inertia. In the domain of voltage stability, graph convolutional networks, transfer learning, and ensemble methods are being explored to address challenges related to system topology changes, fault-induced instabilities, and voltage control in renewable-rich networks. Derivative-free DRL and deep Q-networks offer robust solutions for voltage stability, including load shedding and reference regulation. Generative adversarial networks, deep CNNs, and federated learning show promise for the effective detection and mitigation of stealthy cyberattacks, even when dealing with incomplete or corrupted data. Spatiotemporal measurements and RL further highlight the adaptability of deep learning in safeguarding power grids from evolving threats. Similarly, CNNs, LSTMs, and autoencoders integrated with blockchain and software-defined networking have demonstrated significant potential for cyber-physical intrusion detection.

However, the deployment of these techniques requires overcoming challenges such as data scarcity, scalability, interpretability, and robustness to adversarial conditions. Integration of domain knowledge, development of standardized datasets and benchmarks, and incorporation of explainable AI methods are essential to fully realize the potential of DLTs in power systems. Future research should focus on interdisciplinary collaboration, leveraging advancements in related fields such as cybersecurity, quantum computing, and edge computing to address emerging threats and operational demands. The proactive development of adaptive, scalable, and operator-friendly AI models will be critical to ensuring the cybersecurity and stability of power systems in the face of evolving power systems' landscapes. Pursuing integrated DLT frameworks that address both stability and cybersecurity simultaneously will enable next-generation DLTs' applications to move beyond isolated optimizations, delivering unified, explainable, and scalable intelligence for the secure, stable operation of complex, interconnected, cyber-dependent power systems. By addressing these challenges and exploring the future directions described, researchers and practitioners can harness the power of DLTs to build smarter, more secure, and more sustainable power systems.

CRediT authorship contribution statement

Abdullahi Oboh Muhammed: Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Mohamed Shawki El Moursi:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Nikos Hatziargyriou:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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